Independent Verification and Operation Counts
Supplement to the OBST Research Record

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This supplement preserves the experimental record separated from the focused Monge paper. It contains finite correctness checks, operation counts, search outcomes and implementation qualifications. Its numerical observations do not supply asymptotic proofs.

References to theorems, lemmas, corollaries, equations and sections in the research record retain their checkpoint-71 numbering. The complete checkpoint-71 research notebook and its TeX source are archived beside this supplement. References internal to this supplement use its own numbering. Source files and raw records remain in the repository's `obst/code` and `obst/results` directories.

1 Independent verification and operation counts

The theoretical statements in the paper and archived notebook have proofs independent of these finite tests. Independent cubic recurrences, literal tree enumeration, direct original-cost evaluation, and operation counters provide falsification controls.

The per-outcome overlap audit checks 766 padded dyadic partitions, 4508 scales, and 1,891,752 position/scale incidences, including sizes immediately below and above powers of two. An independent all-root recurrence checks 290 mixed-weight inputs: 1977 original/padded layer equalities, 452,644 local residual bounds, and 28,013 weighted tiles. It examines 685,551 dense diagnostic states and 8,636,426 root candidates. The tests include all binary inputs through three keys, zero mass, uniform and random mixed weights, geometric weights, and mass at either endpoint gap. The executed source and operation counts are recorded in `aggregate_core_audit.json`. These dense verification operations are separate from the sparse construction's theoretical work bound.

Reproduction code and source-hashed records are available in the accompanying repository. The filenames below identify those records; they are not definitions needed by the proofs.

1.1 Implemented condensed constructors

The compact path is now implemented in Python and independently in C++. The product algorithm is the published compression, balanced core split, staircase merge, and decompression method of [2]. We credit that primitive separately from the OBST reduction. A matrix stores its top row, left column, and positive density triples. Two linear bucket passes canonicalize the triples. Boundary extraction uses one-dimensional sweeps; the product merge uses adjacent-entry recomputation along its staircase. Each relevant core row or column is scanned only a constant number of times per merge. None of these operations enumerates the full interval table.

Exact early stopping. The constructors compare consecutive completed layers by their top rows and canonical core triples; the left column is fixed. Equality is therefore equality of the whole table. A fixed point satisfies the ordinary OBST recurrence with zero diagonal. Induction on interval length proves that this recurrence has a unique solution, namely C. Stopping at such a fixed point is exact. Equality of the single full-interval cost is not used. Retained layer witnesses reconstruct a complete optimum: any unfinished nonempty terminal would contribute at least $M > HW$, exceeding the cost of an optimal tree of height at most H. All charges are nonnegative. Reconstruction uses an explicit stack and visits every original key.

Implementation costs and arithmetic. Both implementations retain ordinary integer coordinate maps and recursive witness nodes. Their witness storage is bounded by $O(H(n+K)\log(n+1))$ words, where $K = O(n + \min\{n^2, n\sqrt{WL}\})$ bounds the completed cores. This can exceed the succinct storage bound in Theorem 1 by a logarithmic factor. Compression bounds recursive dimensions by core size, and the core count halves along a recursive branch, so the time bound is preserved. No succinct rank/select implementation is claimed.

The selected perturbation sweep certified amplitude 8 around 1,000 at $n = 256$, amplitude 4 at $n = 512$, and amplitude 1 at $n = 1,024$; larger tested amplitudes at those sizes failed. These are seeded observations, not universal thresholds. Ordinary random inputs failed the certificate. Sizes 384, 768, and 1,000 deliberately skip the shortcut because their minimum-I shape space exceeds the logarithmic width budget.

1.2 Exact bounded-excess construction by dyadic intervals

Validation, failed precursor, and prior work. The implementation passed 1,200 restricted-optimum comparisons against all shapes through eight keys with budgets $0, \ldots, 4$, and 3,000 full mixed-weight checks against cubic DP. It separately checks direct tree costs, both residual-bound implementations, and exact fallback. An additional 104 family checks include zeros, geometric weights, spikes, and key-only/gap-only controls. The seeded size sweep is retained. A precursor substituting ordered depth bounds into the old certificate passed 480 exhaustive checks but rescued none of 24 near-uniform fixtures at its 32-iteration budget. This negative experiment did not prove optimal multiplier search useless; its cost and the failed perfect-tree hypothesis motivated expanding the candidate class.

On the 1,023-key periodic fixture with weights 16 and 17, the excess-two DP finds cost 312,154 and excludes other shapes with bound 312,185. It uses 27,570 states, 137,835 transition checks, 6,052 sorting comparisons, and 1,022 merge comparisons, versus Knuth's 523,776 states and 1,046,532 root candidates. At $n = 1,024$, a seeded amplitude-four perturbation uses 48,003 states and 240,000 transition checks, plus 13,784 sorting and 1,264 merge comparisons. It certifies cost 19,477,996.

The dyadic/prefix-code viewpoint and sequential frontier DP are established ideas. Becker [1] gives a sequential mixed-weight DP for a near-minimal *height* restriction in quadratic time. The band above uses excess *total unweighted path length*, a different restriction; it does not improve Becker's stated problem. Priority of this particular potential, band bound, and exclusion combination remains unestablished.

1.3 A split-height construction covering every input size

Exact verification and operation evidence. The harness passed 450 exhaustive weighted height-class checks through eight keys, 44 exhaustive tall-path minima, 2,000 cubic-checked mixed

inputs with complete fallback (917 successful certificates), and 104 additional adversarial-family inputs. An independent all-root height recurrence checked 520 candidates and 261,482 stored short-table entries; the same 520 cases also compared SMAWK with a retained divide-and-conquer matrix search. The larger sweep has 36 fixtures at three height budgets each. Every near-uniform fixture certified, including previously skipped sizes 384, 768, 1,000, and 1,536. All random, geometric, and gap-spike fixtures failed the certificate, sometimes despite an optimal restricted candidate.

At $n = 1{,}000$, the minimum-height candidate and full optimum are both 18,973,413, below the exclusion bound 18,974,193. The optimizer uses 117,453 short states, 220,483 short root candidates, 84,937 genuine matrix queries, 10,268 padding queries, 16,016 prefix scans, and 5,630 coarse transitions. The bound adds 10,648 sorting and 1,335 merge comparisons, plus 1,000 contractions; all keys are reconstructed. Knuth uses 500,500 states and 1,002,162 root candidates. These distinct operation types are reported separately, without treating them as equal-cost instructions.

1.4 A variable interval frontier for exact full construction

Independent verification. The harness passed 960 weighted exhaustive-shape comparisons through eight keys on arbitrary nondecreasing contours, including infeasible classes. It checked 914 interval costs independently and 90,193 top minima by brute search. Another 2,400 full solves, using both modes on 1,200 inputs through 40 keys, matched cubic DP and reconstructed costs. All 24,600 internal subtrees in the cubic-optimal trees satisfied Lemma 2. Families include zeros, ties, geometric variants, spikes, and key-only and gap-only controls. A further 56 fixtures through 2,048 keys matched Knuth in both modes, including ordinary random inputs that failed the preceding height certificates.

At 2,048 random keys, the adaptive version chose depth six and returned cost 21,053,524. It used 309,261 interval cells and 616,642 root candidates, versus Knuth's 2,098,176 cells and 4,215,527 candidates. Additional work included 24,035 contour mass checks across seven parameter depths, 1,245,431 genuine matrix queries, 149,319 padding queries, 131,136 scans of each row/column type, and 107,859 coarse transitions. The uniform version stored 337,550 cells. On log-uniform weights the corresponding cell counts were 308,487 versus 431,951; on a single heavy key they were 1,048,576 versus 1,573,376. Geometric inputs selected the full table. All operation categories and parameter levels are retained separately.

1.5 Dominant-gap separators and local mass frontiers

Verification and complete work accounting. The harness exhaustively checks 22,140 weight assignments in $\{0, 1, 2\}$ through four keys against every tree shape. It checks 3,211 qualifying gap certificates and their cost identities, 2,036 endpoint certificates, and 380 key certificates. A further 3,280 full solves, using both frontier and Knuth terminal strategies on 1,640 inputs through 40 keys, match independent cubic costs and direct tree evaluation. Inputs include empty, zero, tied, geometric, spike, key-only and gap-only controls. Both strategies and the direct global frontier also match Knuth on 40 fixtures through 2,048 keys.

At $n = 2048$, the direct and local frontier interval counts are:

Fixture	direct frontier	local frontier	terminal components
Random	308,115	308,115	1
Near-uniform	309,290	309,290	1
Geometric	2,098,176	0	0
Heavy key	1,048,576	153,660	2
Heavy gap	1,047,552	153,582	2
Random background, heavy gap	1,047,552	153,362	2
Hierarchical gaps	522,242	22,333	64

Cell counts do not include all work. On hierarchical gaps, 63 gap separators require 254 endpoint checks, 6,142 index-build comparisons and 1,476 range query comparisons. Terminal solves use 42,885 root candidates, 39,284 genuine top-matrix queries, 6,261 padding queries, 8,122 contour mass checks, 6,952 scans of each row/column type, and 1,792 coarse transitions. Their 1,922 keys are reconstructed and relabeled. On the heavy-gap fixture, 305,118 terminal root candidates and 1,214,092 genuine top queries accompany the reduced interval table. All counter categories are retained; metadata such as local selected depths and actual heights are summed over terminals, not reported as a global tree height. Geometric input uses 2,048 endpoint separators and 4,095 endpoint checks, with no index or terminal work.

1.6 Joint heavy-outcome cores

Independent checks and operation counts. Every nonempty selected-outcome subset and every tree shape through four keys were examined. In all 1,937 non-core-closed cases a core-closed tree has componentwise no-greater selected depths and at least one strict improvement. On 22,140 inputs with weights in $\{0, 1, 100\}$, all 17,054 qualifying nontrivial certificates match the optimum over every shape and the composition identity. Another 2,640 full solves on 1,320 inputs through 32 keys, with both kernels, match independent cubic DP and reconstructed costs. On 520 small inputs the threshold sweep matches direct enumeration of every threshold and payload partition. Both kernels also match Knuth and the previous separator constructor on 30 larger fixtures through 2,048 keys.

At $n = 2048$, the following cell counts include both core and payload solves, or the complete fallback when certification fails:

Fixture	previous separator	joint core/frontier	core keys
Random	309,201	309,201	—
Near-uniform	309,290	309,290	—
Two heavy gaps, unit background	2,098,176	10	4
Two heavy gaps, random background	2,098,176	306,506	4
45 heavy gaps	304,486	23,965	90
45 moderate gaps	304,486	23,695	90
45 mixed heavy outcomes	305,421	24,016	68

The moderate-gap fixture sets each selected weight to exactly $(H + L)W_0 = 42{,}395{,}178$, with $H = 9$, $L = 12$, and $W_0 = 2{,}018{,}818$. The old size-based condition would require $(90 + 12)W_0$ and fails. Work includes 63,549 sorting comparisons, 2,047 Fibonacci additions, 3,024 height checks, 1,958 activations, 1,912 unions, and 1,511 find steps. Kernels use 45,365 root candidates, 116,338 genuine top queries, 23,904 padding queries, 12,336 contour checks, and 17,488 scans each for rows and columns. All 2,048 keys are reconstructed. Local kernel metadata such as heights and chosen depths are sums across calls, not global tree heights. The uniform two-gap case still pays selection, 2,044 payload-node constructions and relabelings, and ten core DP cells; it does not perform only ten total operations.

1.7 Joint heavy-outcome cores

Full construction beyond the quadratic-table scale. On the unit-background two-heavy-gap family, full trees were constructed at $n = 512, 2048, 8192, 32768, 65536$. Each exact cost matched an independent cubic solve of the four-key instance

$$p = (1, 1, 1, 1), \qquad q = (1, n^3, 2n - 7, n^3, 1),$$

plus $2I_*(n-4) + 3(n-4)$ for the uniform payload, and direct evaluation of the full reconstructed tree. The uniform closed form was separately checked against cubic DP through 16 keys. At $n = 65536$, the exact cost is 1,407,374,885,846,941. The construction used ten core DP cells and fifteen core root candidates. It also used 196,799 sorting comparisons, 65,535 Fibonacci additions, 65,535 height checks, 65,532 activations, 65,531 unions, 65,529 find steps, and 196,609 uniformity comparisons. All 65,532 payload keys were constructed and relabeled, and the four core keys were reconstructed.

1.8 Residual-cost certificates for heavy-outcome cores

Independent exact checks. On all 22,140 assignments in $\{0, 1, 4\}$ through four keys, every one of 17,054 nontrivial threshold cases passed the excluded-class lower-bound check against all tree shapes. There were 13,669 passing witness tests and 15,181 passing exact-candidate tests, including 1,512 cases rescued by refinement. Another 14,912 excluded-tree checks used arbitrary selected subsets, weights in $\{1, 3\}$, and up to three keys. These latter selections need not be weight thresholds.

Both kernel choices and both refinement settings passed 6,560 full solves on 1,640 inputs through 40 keys against independent cubic DP and direct tree evaluation. All 1,640 candidate portfolios matched direct enumeration of the two representatives in each core-size band. Forty-eight larger fixtures and nine additional fixtures with independently varying positive key and gap weights matched Knuth through 2,048 keys. The varying backgrounds are integers in $[1, 1000]$ and their selected outcomes have weight $4000n$; the unit-background success rows below use weight $4n$. These are finite tests, not an assertion that every such random instance passes certification.

At $n = 2048$, selected operation categories are:

Fixture	Previous intervals	New intervals	Protected proof cells
Two linear-weight gaps, unit background	568,350	10	10
45 linear-weight gaps, unit background	304,486	1,394	4,095
45 mixed outcomes, unit background	305,421	1,250	2,346
Two gaps, varying background	951,851	307,541	30,941
45 gaps, varying background	304,486	23,757	22,519
45 mixed outcomes, varying background	305,421	24,163	27,558
Random control	308,860	308,860	33,380

New intervals count all executed full-weight kernel cells, including fallback, but exclude the separately displayed protected-core proof cells. The proof column sums all attempted cores. Selected-state metadata is not counted again as executed cells. Sorting, profile scans, lower-bound contractions, witness construction, kernel matrix work, and reconstruction have separate raw counters; this table is not a total instruction count. The random control fails all ten witnesses and pays additional proof work before fallback.

1.9 Residual-cost certificates for heavy-outcome cores

Full large constructions and a remaining certificate gap. For every $n = 2^h$, $3 \le h \le 16$, the full $\theta = 4n$ construction matches Equation (120), direct evaluation of the entire output tree, and an independent four-key cubic core plus the uniform payload formula. At $n = 65,536$, the optimum is $3,604,381$. The proof core and full-weight core each use ten cells and fifteen root candidates. Other counts include 196,799 selection-sort comparisons, 131,204 bound-sort comparisons, 65,536 bound contractions, 65,535 two-queue comparisons, 65,532 activations, 65,531 unions, and 65,529 find steps. All 65,536 witness keys are constructed and evaluated, then all output payload and core keys are constructed. The lower bound is $3,604,482$, giving a certificate margin of 101. The previous method was only planned at this size: it rejects the core and selects 146,498,625 frontier cells at depth 11.

Failure remains informative. At $n = 2048$ and selected gap weight $2n$, the exact core-closed cost $71,611$ equals the unrestricted optimum, but the exclusion lower bound is only $67,586$; even refinement cannot certify it. At weight n, the closed candidate costs $61,371$ while the unrestricted optimum is $59,332$. Thus some failures reflect a weak certificate and others reflect a genuinely suboptimal proposed structure. Improving the bound must preserve this distinction.

Raw results and source hashes are in `core_exclusion*.json` under `obst/results`. An initial full-harness failure was a metadata bug: list-valued fallback fields were passed to arithmetic counter update. Direct assignment fixed it, and the final full harness was rerun. A completed single-threshold experiment was superseded after it missed several useful cores; the published selector and every final result use the bounded portfolio. The general subquadratic objective remains unresolved.

1.10 Coupled two-gap skeleton certificates

Verification and measured limits Independent forest enumeration passed 11,880 checks. Exhaustive small weight inputs numbered 24,188, with 51,654 pair-exclusion checks and separate checks for both first and tail lower bounds. Among these are 3,288 exact closed candidates accepted by the coupled bound but rejected by B_{basic}. Complete construction passed 6,560 comparisons with independent cubic DP, covering both payload kernels and refinement modes. Larger tests at $n = 128, 512, 2048$ include random, near-uniform, geometric, single-spike, adjacent-pair, interior-pair, and varying-weight inputs. Neither favorable examples nor these exhaustive finite checks replace the preceding proofs. Raw counts and source hashes are stored in `pair_skeleton*.json`.

The nonuniform-family harness checks 31 placement/size combinations, including clustered, spread, and random placements; 28 are compared with Knuth and eight with cubic DP. All satisfy the exact residual formula and the proved certificate margins. Fourteen unit-background cases extend through $n = 65,536$. At $n = 2048$ with unit keys and selected gap weight $2n$, the earlier bound $67,586$ is replaced by $71,619$, which certifies the exact cost $71,611$. At gap weight n, the closed candidate remains suboptimal and is correctly rejected.

At $n = 16,384$ the seeded nonuniform instance has 1,023 doubled key weights. Full construction returns cost $767,240$, with witness $768,946$, first bound $780,208$, and tail bound $795,651$. Its 7,758,912 short states, 3,922,465 top states, and 20 total core states give 11,681,397 DP states versus 134,225,920 in a full Knuth table. Parameter-selection metadata is not counted a second time. There are also 42,351,013 top-matrix queries, 15,502,473 short-root candidates, 261,181 forest sorting comparisons, 49,143 forest merges, and 256,673 contour mass checks. The complete tree is reconstructed and its cost evaluated directly. The largest instance is certified by the theorem and construction, not compared with an independently executed quadratic table.

The chief unresolved issue is how to retain useful path-union information when no constant-size

core can be certified. The present method does not bound the number of necessary boundary states for arbitrary mixed input. Its general worst-case construction remains quadratic.

1.11 An ordered depth-chain lower bound

Preliminary strength and cost. At $n = 256$, normalized gaps $(\mathrm{OPT} - \mathrm{LB})/W$ for the chain bound versus order-free Huffman were 0.15003 versus 0.22080 on a random mixed instance, and 0.00873 versus 0.49294 on an alternating instance. However, 300 ascent iterations used 1,082,368 key-depth cells including initialization, compared with 32,896 interval states for Knuth. The unique-root variant uses two passes and still leaves 254 of 256 potentially improving roots on the random instance, and 255 on the alternating instance. Only two survive on the gap-spike instance. These results identify a lower-bound component and its limitations; they do not establish a faster general exact solver.

1.12 Exact depth ordering and an arbitrary-window obstruction

Independent verification. The harness enumerated all 873,612 sequences in $\{1, \ldots, n\}^n$ for $1 \leq n \leq 7$. The characterization accepted exactly the independently generated BST depth sequences: respectively 1, 2, 5, 14, 42, 132, and 429. The stack recognizer agreed with direct interval scans on every sequence. Width3 and exact flow already admit four invalid sequences at $n = 6$ and 34 at $n = 7$. The positive family was checked at $t = 16$ for all 127 widths $2 \leq w \leq 128$: exact costs, window validity, zero flow, and the tree-cost lower bound all passed. Knuth and direct tree evaluation checked every instance; independent cubic DP also checked $w \leq 24$. Another 120 cases combine $2 \leq w \leq 16$ with forced prefix depths $1 \leq D \leq 8$. Cubic DP, Knuth, direct costs, and inspection of the optimal prefix verify the exact additive identity and preserved strict gap.

1.13 Endpoint responses: insufficiency and representation size

Independent checks and remaining question. The exact-arithmetic harness checks 875 response queries against cubic DP on 175 inputs through seven keys. Rational half-plane clipping checks exposed pieces; 725 retained planes with positive endpoint weights satisfy the Fibonacci bound. The geometric family passes 408 hinge queries, including thresholds and neighboring integers; 1,496 two-parameter probes through 33 keys; 204 full cubic comparisons; 30 all-shape checks that every optimum has the prescribed depth pair; and 680 strict endpoint inequalities. The product checks execute 887,808 Knuth root candidates. Another 304 queries expose all outer-root pieces, and 384 queries test arbitrary parameters and the global transition threshold. All 2,184 factored complete constructions agree with Knuth and direct evaluation, and construct exactly n keys. Full-quadrant envelopes through nine keys have 5, 10, 17, and 26 pieces, agreeing with the proved formula. The factored constructions use 14,492 threshold comparisons, 8,736 outer-minimum comparisons, and construct 52,232 keys in total; these categories exclude the independently executed reference solvers.

The collision search found an initial surrounding context with outer weights 16; reducing them to seven preserves the one-unit optimum difference. The published certificate independently checks domination of discarded planes and enumerates both full eight-key tree sets. It does not depend on the clipping implementation or sampled parameters. Code, raw results, and the harness source hash are in `boundary_response.py/json`.

Legal-domain collision and signed recovery. The independent recovery audit checks 800 integer inputs: all binary mixed vectors through four keys and seeded inputs through 24 keys,

including geometric, uniform, gap-only, key-only, and zero weights. It compares 3968 signed value queries against a fresh all-root recurrence using 1,926,832 root candidates, with negative endpoint weights permitted. Literal shape evaluation supplies 107,608 further signed cost controls. The boundary-path classification has 19,184 shape controls and 2792 unique-chain checks. All recovered gap vectors agree exactly, and each instance checks the query-count and rational-denominator bounds.

The whole legal quadrant is certified by two complete intercept tables and 18 exact domination checks. Independent enumeration evaluates all 2860 ordered trees across the two eight-key contexts and agrees with the root-conditioned costs. The signed absolute probe $(0, -1)$ is checked by the original recurrence. Twelve larger recoveries reach 65,536 keys using the separately proved chain-line oracle. Random and uniform instances at that size use 65,536 value queries each; the all-zero instance uses three. These larger rows verify complete recovery from the chain representation; they have no independent large signed-DP oracle. The result record retains counts, the exact collision, verification scopes, and five source hashes. The recovery theorem bounds calls to its specified oracle and does not provide a faster general OBST solver.

1.14 A core-sparse Monge reduction with terminal penalties

Independent evidence. The condensed matrix harness passed 1,600 random rectangular products, checking 251,779 output values and exact leftmost witnesses by enumeration, and 1,600 arbitrary submatrix selections. Sparse distance-matrix products at dimensions 128, 512, and 2048 each retain seven nonzero output core entries from seven entries in each input. Core visits remain 673 in these three checks, while dimension scans grow. This exercises nontrivial sparse output, not just products with empty cores.

The complete Python constructor passed 1,134 solves, including 682 exhaustive binary-weight cases through four keys (including two empty-key inputs), 838 cubic checks, and 32 fixed-budget controls that disable early stopping. On 78 layers of the small controls, all 3,367 interval values and leftmost witnesses agree with explicit all-root enumeration. Full tree audits check 15,544 reconstructed keys. Geometric and folded weights through 64 keys remain adversarial controls; the geometric input requires all 64 layers.

The C++ address/undefined-behavior sanitizer run passed 1,278 complete solves, including 826 independent cubic checks, 12 Python compact comparisons, random weights through 48 keys, geometric/folded inputs through 64 keys, and all-zero and single-active-gap controls. It checked 18,248 reconstructed keys. The release run passed 1,362 solves and 88,904 reconstructed keys, including 1,302 independent Python Knuth comparisons. All release cases also compare against the separately implemented C++ Knuth constructor. The independent parent-array audit verifies inorder validity, unique key coverage, and direct mixed weighted cost. Three malformed/out-of-domain inputs are explicitly rejected by each binary. No sanitizer error was reported. Results include source hashes, compiler flags, and seeds.

1.15 A centered core bound for large integer baselines

Independent tests and remaining limitations The dense falsification harness `centered_layers.py` passed 1,100 complete solves, 824 cubic comparisons, 3,317 all-root cells, 1,538 baseline all-shape checks, 99,087 baseline-density checks, 685,288 tile checks, and 2,327 scale checks. It explicitly computed 4,153,053 dense states and 8,223,340 root candidates and reconstructed 11,656 keys. Baselines include 2^{256} and 2^{200} independently for keys and gaps. Zeros, concentrated gap mass, and geometric weights remain in the suite.

At 256 keys, three large-baseline fixtures have deviations 501, 505, and 504, normalized height budgets 14, 14, and 12, and maximum interior cores 6,428, 6,386, and 6,351. The earlier numerical-mass budgets were all 256. These are budget comparisons, not measured speedups: the original constructor already stops at exact fixed points. The 64-key geometric control still has $H = 64$ and all 2,016 interior density cells nonzero.

The actual condensed variant `centered_obst.py` passed 999 complete solves, 790 cubic comparisons, 32 fixed-budget controls, 1,001 intermediate values and leftmost witnesses, and 12 original-compact controls. It checked 9,992 reconstructed keys. Its large-baseline 256-key fixtures finish in ten layers with completed cores 6,624, 6,540, and 6,496. The earlier solver and its source-hashed compiled performance controls remain unchanged.

1.16 Ruler patterns and a dense-core representation barrier

Exact checks, counts, and controls The dense `ruler_layers.py` diagnostic checked 921 layer solves, 70,100 weighted-subsequence interval identities, 22,437 range maxima, 20,432 all-root cells, and 2,168 all-shape layers. It explicitly computed 5,708,220 layer states and 11,108,348 candi-

dates. Its separate full-table controls used 1,422,592 states and 2,692,718 candidates, with 96 cubic comparisons. These are diagnostic costs, not an implementation of a new fast layer constructor.

The family audit checked 68 sizes through 512, 221,284 interval/formula equalities, 202,127 instances of (143), 179,486 instances of (144), 2,974 enumerated shapes, and 13 cubic controls. At $n = 512$, all 130,816 ordinary core cells and all 129,795 valid fourth-difference cells were nonzero. The theorem uses only its explicitly proved quadratic subset. Formula and direct tree-cost checks continue through 65,536 keys without dense interval tables.

The full exchange/gate/fallback constructor passed 1,210 solves, 826 cubic comparisons, 658 checks against the prior minimum-path-length shape DP, and 386 independent enumeration checks over 950 minimum-path-length shapes. It certified 349 inputs; 512 failed the exchange condition and 349 passed that condition but failed the global gate. All fallback costs were checked. Both orientations and nonmonotone compensated key fixtures are included, alongside ordinary random and geometric failures.

1.17 Exact transfers for joint key/gap signatures

Independent checks and complete work counts The identity harness preceded the implementation. It checked all forty boundary-count intervals through block length ten, 1,365 direct costs using independent slot-expanded tree shapes, and 228 block path-independence identities. All 120 feasible short boundary/count reconstructions were checked. Another 1,000 forward/backward transfer vectors were compared with explicit word enumeration, including negative profile values and unreachable entries. The oracle does not reuse the production deque.

The final source-hashed M4 suite passes 1,023 standalone full solves, 778 cubic comparisons, 1,023 unrestricted shape-DP controls, 762 independent shape enumerations, and 1,023 independent checks of multiplicity sorting. There are 366 certificates and 657 gate rejections, including 411 in preflight. Another 1,023 portfolio solves return the same exact costs: 912 select signatures and 111 select the affine route. Eight large portfolio controls agree with the standalone constructor. Inputs include zeros, ties, large integers, geometric weights, arbitrary block values, and one-key blocks. Both scaling families at 1536 keys also match Knuth and the unrestricted shape optimizer.

Table 1 includes every block transfer performed during traceback, not just the initial forward sweep. The four-block 98,304-key solve also scans 2,228,224 source positions, pushes 311,317 deque entries, and performs 311,283 deque value comparisons, 262,146 traceback split checks, 98,304 residual contractions, and 110,590 bound-queue comparisons. Its certificate sorts thirteen multiplicity records using 26 comparisons and expands all 196,609 input weights. The sixteen-block solve uses forty-nine records, 167 sorting comparisons, 2,207,873 deque pushes, 2,415,139 deque value comparisons, and 524,310 traceback split checks. Both construct and directly evaluate all 98,304 keys. These are distinct operation categories, not a machine-instruction total. Maximum profile width counts ranks and is not a measured peak-memory byte count.

1.18 Global alternating-gap bounds and linear profile construction

Independent evidence and full-construction counts The exact word/tree harness preceded the production constructor. Recognition matches literal word enumeration on all 1,022 nonempty binary patterns through nine keys; 865 eligible patterns additionally match independent minimum-path-length tree enumeration. A separate exhaustive oracle checks 13,834 parity/tree inequalities and 69,170 mixed-weight/tree inequalities through nine keys, including exact equality of each size/parity minimum with the gap formula. The perfect-size counterexample is retained.

The full suite passes 3,157 standalone solves against Knuth, including 2,847 cubic controls and

Keys	Blocks	Window queries	Potential terms	Max. profile width
1,536	4	40,968	10,242	1,024
1,536	16	188,552	47,138	1,024
6,144	4	163,848	40,962	4,096
6,144	16	753,800	188,450	4,096
24,576	4	655,368	163,842	16,384
24,576	16	3,014,792	753,698	16,384
98,304	4	2,621,448	655,362	65,536
98,304	16	12,058,760	3,014,690	65,536

Table 1: Actual full-constructor transfer counts on certified non-affine mixed signatures, including linear-space traceback recomputation. Larger-than-1536 cases use the proved certificate and direct output-cost checks, rather than executing the dense comparison tables.

1,325 class-candidate comparisons against the unrestricted shape DP. Another 3,157 routed solves agree. There are 1,201 accepted full solves: 1,083 use the global profile equality, 116 use the older mixed gate on perfect shapes, and two are empty instances. The remaining 1,956 route to exact fallback: 86 have unsupported key alphabets, 600 unsupported gaps, 241 unsupported high counts, 905 fail profile attainment, and 124 fail the perfect-shape mixed gate. Zeros, ties, huge integers, random, geometric, near-uniform, arbitrary binary patterns with and without a baseline, and earlier affine inputs are included.

Both scaling families now have no added baseline. At 393,216 keys, a nonperiodic witness-generated pattern has 262,111 joint-signature runs but only 196,305 interval transitions; strict alternation has 393,216 runs and 196,608 transitions. They require 212,793 and 262,144 membership queries respectively, and one traceback step per stored high boundary. Each orders two multiplicity records, expands 393,216 key weights, constructs every key, and directly checks the full mixed cost. Neither uses a residual-Huffman contraction or a dense shape state. These are operation categories, not machine-instruction totals or measured peak bytes. Ten scaling portfolio controls agree; the two 1536-key cases also match Knuth and the unrestricted shape DP. Larger cases use the proved global bound and direct output-cost verification.

1.19 Threshold runs and a linear Cartesian certificate

Independent evidence and complete operation counts The threshold probe preceded the production constructor. Across all 1022 binary gap patterns through eight keys it checks 862118 indicator/tree inequalities and equality with the exhaustive optimum for every pattern. It checks 20560 weighted tree inequalities and key equality characterizations, plus 1200 mixed cubic-DP controls.

The full constructor harness checks 3000 cases against independent cubic and Knuth recurrences, including 2610 literal exhaustive controls, and another 3000 portfolio solves. The Cartesian attempt accepts 885, rejects 927 key profiles, and rejects 1188 gap profiles. The harness also checks 21844 gap vectors against explicit threshold and sorted-rank sums, 36 positive multilevel fixtures against Knuth, six larger family instances, and a 20000-key linear-height tree. Twenty independent selection scaling controls cover random, ascending, descending, duplicate, and organ-pipe weights through 100000 gaps.

At $n = 327679$, there are 327679 distinct key weights and 65538 gap levels. The exact full cost is 253331707448885247. The attempt uses 655322 Cartesian comparisons, 327679 key visits, 327680 gap visits, 327680 gap-bound scan items, and reconstructs all 327679 keys. The checked

extrema formula avoids selection on this family. There are no interval-DP cells. For the general selection controls at $N = 100000$, the sum of outer rank-level sizes is 231070; all recursive selection, partition, copy, sum, and comparison counts are separately retained in `threshold_complete.json`. These counters describe logical work, not measured peak bytes or machine instructions.

1.20 Shifted alphabetic certificates beyond key-capacity equality

Independent checks, scaling, and limitations The independent probe precedes the optimized constructor: 960 cubic controls, 3629 scalar-transfer checks, and 37500 literal tree inequalities. It certifies 138 candidates, including 24 strictly above the old bound. The constructor checks all 9840 ternary gap vectors through seven keys against cubic values and the literal compatible-pair implementation, including identical ordered gap depths. Another 960 full mixed solves match cubic and Knuth, including 560 exhaustive controls: 336 certificates and 624 fallbacks, with all 279 constant-key cases accepted. Random, near-uniform, geometric, binary, and zero inputs are included. Eight family sizes through 895 keys match Knuth; the four smallest also match cubic. A 1199-key skewed tree tests iterative traversal.

Five large constructions reach $n = 229375$, with 196608 distinct key weights, capacity deficit 32768, and exact full cost 2029411090432. The independent explicit family tree, constructed tree, and bound agree. The largest run uses 229375 compatible merges, 458749 priority pushes and pops, 6885341 priority comparisons, 5136371 heap-item comparisons, 1473827 key-sort comparisons, and 5021690 meld calls. It reconstructs every key and uses no interval-DP cells. The solver receives only p, q.

1.21 Endpoint prices and the limit of additive transfers

Independent evidence and measured costs The independent harness precedes the optimizer. It checks 2052 literal transfer identities and exact cost-difference ranks through eight keys: rank $2n - 2$, nullity three. For 60 three- and four-key instances, an independent arrangement of all alphabetic-tree and key-rank breakpoints evaluates 5111 rational points; every maximum agrees with the price search. On 600 mixed cases through 64 keys, 1712 individual oracle values match independent cubic values, and small cases also match literal trees. All 600 bounded full solves match cubic, Knuth, and direct cost: 220 certify, including 152 previous rejections, and 380 fall back. The uncapped diagnostic certifies 228, rescues 160, and completes every maximum within ten calls in this suite. There are 358 strict bound gaps and 600 clamped-transfer monotonicity controls.

Nineteen exact reduction controls produce the four-key obstruction. A further 6000 seeded positive cases find no nonintegral price maximum; this is not an integrality theorem. Intermediate prices are fractional: the obstruction search queries $y = 1/3$ and obtains $F = 68/3$. The code uses exact rationals and does not assume integral maxima.

1.22 Root/nonroot coupling beyond the transfer barrier

Independent controls, regressions, and a retained conjecture The independent case-partition harness precedes the optimized constructor. Across 840 mixed instances through eight keys, it checks 246480 literal conditioned inequalities. The bound strictly improves $K + A$ on 373 cases and is exact on 324. Random, near-uniform, geometric, and zero/binary inputs are included.

The normalized full implementation matches cubic, Knuth, and reconstructed original cost on 1040 instances through 64 keys; 720 also match literal tree enumeration. There are 800 independent bound controls and six additional family checks. The route certifies 505, including all 240 constant-size base cases, and falls back on 535. It exceeds the completed maximum transfer bound in 84 cases and rescues 11 price rejections, but loses 47 price certificates. These finite counts are not distributional guarantees, and the fixed endpoint choice is not universally superior to price optimization.

A separate diagnostic retains an unproved marked-boundary event family. It inserts a barrier at one weighted interior key, permits ordinary compatible merges on either side, and charges the key weight when a pair first consumes that barrier. Greedily taking the cheapest event fails: for $p = (0, 4, 0)$, $q = (8, 10, 2, 18)$, it returns 82 instead of 80, with strict primary choices. Inverting a unit uniform transfer gives the positive instance $p = (1, 5, 1)$, $q = (7, 9, 1, 17)$, with costs 79 and 77 respectively.

The alternative probe tries the cheapest crossing pair after every prefix of ordinary merges, completes compatible merging, and evaluates reconstructed trees with the original positional key cost. It matches all 17769 checked instances, including 15769 literal-tree controls; 4134 controls also enumerate every crossing choice without changing the best result. This is a conjectural candidate family, not an exact solver theorem. There are 20800 invalid depth reconstructions across the attempts, which are rejected. Merge charges are not assumed to equal positional costs.

Its candidate costs are not unimodal in the activation prefix. With gaps $(17, 83, 93, 11)$ and weight 92 on key 1, the ordinary merges cost 100 and 104; activation before, between, or after them gives costs 500,681,500. Thus a single-valley search is unjustified even with positive gaps and strict ordinary priorities. The full probe records 6759 nonunimodal sequences, 116 greedy failures, and 65 charge/cost disagreements. No subquadratic complexity is claimed for the probe. The proved root/nonroot certificate does not depend on this conjecture.

1.23 Exact checks and preliminary measurements

The code uses nonnegative integer weights, an independent cubic recurrence, Knuth's recurrence, and direct evaluation of reconstructed trees. The basic harness additionally enumerates all tree shapes for small sizes. CPU experiments ran on the sponsor's M4 Mac mini. The reported quantities are algorithmic counts, not inferred asymptotic bounds from elapsed time.

- Single-key contraction: 22,143 exhaustive weight assignments in $\{0, 1, 2\}$ through $n = 4$, and 6,000 seeded random instances through $n = 20$. Every cost and reconstructed tree matched the cubic oracle.
- Block theorem: 19,486 qualifying interval contractions across 4,500 seeded instances through $n = 16$ matched independent cubic costs.
- Block algorithm: 3,000 random instances through $n = 20$, including fixed and adaptive thresholds, matched the cubic oracle and tree evaluation. Additional small cases check threshold selection against every threshold.

In the heavy-separator instance all gaps and ordinary keys have integer weights in $\{1, 2, 3\}$, and 22 randomly selected keys have weight 10^6. The block algorithm retains exactly 22 keys. Its state reduction is about 12.7-fold, but this is not a claim of a 12.7-fold runtime speedup. On ordinary random mixed weights the gain is small. The single-key pass contracts every key for the tested sequence $p_i = 4^i, q_i = 4^i$, but not for most of the ratio-two sequence. Geometric integer inputs also illustrate why bit complexity must be distinguished from arithmetic-operation complexity.

1.24 Additional shape and contraction controls

The recursive constructor passed 3,000 random mixed inputs through thirty keys against cubic recurrence and direct tree evaluation. The minimum-path- length optimizer matched literal shape enumeration on 360 weighted inputs through eight keys. Its full fallback variants matched 3,000 further inputs. The affine-corridor audit includes independent candidate, regret, and complete construction checks; its parameter-dependent state bound is proved in the main text. Source-hashed records retain the individual operation categories.

1.25 Additional recurrence and certificate controls

Reference checks and implementation boundary. The dense audit `penalty_layers.py` passed 1,092 complete solves: 680 exhaustive binary-weight inputs through four keys, 384 random inputs through 32 keys, and 28 size/family fixtures through 256 keys. These include zeros, near-uniform, alternating, varying mixed, and geometric weights. All complete values and trees match Knuth and direct cost evaluation. There are 7,972 independent cubic interval checks, 8,048 all-root cell checks, 792 explicit padded-product checks, 792 all-shape truncated-layer checks, and 139,992 verified dyadic tiles. The geometric family correctly needs all 64 layers at 64 keys; no logarithmic-in-n claim is made for its exponentially large mass.

At 256 keys, the maximum interior core over all layers was 3,857 for mixed weights in $\{1, 2\}$, 6,196 for weights in $\{100, 101, 102\}$, and 5,008 for mixed weights in $\{1, \ldots, 16\}$, against 32,640 possible interior positions. These are representation counts, not runtime speedups. The audit actually computed 6,318,862 dense states and examined 12,381,033 root candidates across its full suite. It constructed 11,912 keys. Source-hashed raw data is in `penalty_layers.json`.

The lexicographic oracle matched 400 exhaustive shape-profile comparisons through eight keys. Uncapped search matched exhaustive evaluation of all allowed integer prices on 1,200 inputs. On 3,000 additional full-OBST inputs through thirty keys, both capped fallback versions matched cubic DP and direct tree costs; 87 previously unsuccessful unrestricted certificates were recovered. A separate seeded sweep recovered the $n = 1,024$, amplitude 2 input with two additional oracle calls, and the $n = 2,048$, amplitude 1 input with three. Larger tested perturbations usually exceeded the profile ceiling. This applies standard concave-price optimization to the certificate; it is not a general subquadratic result.

In the seeded 2,048-key case, the candidate costs 43,032,446; the bound rises from 43,031,347 at $\delta = 0$ to 43,032,459 at $\delta = 125$. The three extra passes use 6,144 contractions and 6,141 pair comparisons, in addition to 4,096 shape states, 6,142 transitions, 20,784 sorting comparisons, and 2,752 original merge comparisons. Knuth uses 2,098,176 interval states and 4,198,710 root candidates on that input.

Twelve full constructions reach $n = 262143$, with 262141 distinct key weights, 262144 distinct gap weights, and cost 92530765201355. The best preceding transfer bound is 91912294498251, a deficit of 618470703104. The largest run uses three alphabetic calls, 524285 compatible merges, 13894100 priority comparisons, 17365446 heap-item comparisons, and 1482337 key-sort comparisons. All scans, heap events, rebasing, and output visits are recorded; there are no interval-DP cells. Seven sizes through 511 keys match Knuth and three match cubic. Larger sizes match a separately constructed perfect tree and the closed form. Six smaller sizes additionally complete the exact endpoint price search and verify the full-class deficit independently.

Both oracles passed 50 comparisons against exhaustive length-sequence minimization, 5,000 arbitrary-multiplier bound checks against cubic OBST, and 1,050 direct comparisons of the full and shortened oracles. Forced-root marginals were additionally checked against exact root-constrained costs.

The initial probe checks 1092 alphabetic controls, 2400 mixed controls, 616500 literal tree evaluations, and 42563 pair evaluations: 2385 candidates are exact and 15 are suboptimal. Exhausting key multisets and ordered gaps over $\{0, 1\}$ and $\{1, 2\}$ through seven keys gives 7172 cases, including 14 binary failures and no $\{1, 2\}$ failures; the latter finite observation is not a theorem. There are 2036 constant-key controls, 6918 profile/depth identities, and 1421 transfer controls. All 8192 integer corners of $1000(p, q) \pm 1$ retain a gap at least 996. Padding is checked by Knuth through 256 keys (largest weight 732 bits), with cubic checks through 16.

All 512 integer corners of $100(p, q) \pm 1$ are checked; the smallest observed gap is 94, above the proved lower bound 56. With all nine weights distinct, the example

$$p = (996, 997, 1998, 999), \qquad q = (1000, 2001, 1002, 1003, 1004)$$

has full optimum 23999 and maximal transfer bound 23004. A constant supporting plane certifies that maximum in one call.

1.26 Coarse optimal families and complete fine-weight refinement

The independent audit checks 716 coarse inputs: all 680 binary mixed inputs through four keys, and 36 seeded inputs at $n = 5, 8, 12$. All 7540 interval minimum/maximum optimal-root pairs agree with a separate all-root recurrence. Independent optimal-tree counts agree with every uniqueness decision; 311 coarse optima are unique. The audit then checks 1432 complete signed-rational refinements against the original all-root optimum, original-weight tree traversal, and coarse optimum. Every fine payload cell count equals $\sum_j m_j(m_j + 1)/2$.

Six further controls exercise automatic scale selection; four exercise certificate rejection. Three zero-coarse-mass controls preserve the full optimal family, including optima taller than a balanced sufficient height. Twenty queries reuse four prepared partitions, agree with independent all-root optima, and perform no new Monge products. The record `coarse_face_audit.json` retains its seed, counts and eight executed-source hashes.

The scale audit constructs eight unstructured lattice perturbations. Coarse weights are independently drawn from $\{0, \dots, 7\}$; a large integer scale and distinct position perturbations make all $2n+1$ original weights different. Automatic scale selection accepts at $m = 7$ in each retained row. Complete construction includes both coarse orientations, the unrestricted-child closure products, partition extraction, and every fine payload. Selected exact counts are:

n	Trial	Forced keys	Largest payload	Fine DP cells	Core visits
32	0	32	0	0	553,436
32	1	32	0	0	476,696
64	0	56	3	15	1,910,066
64	1	62	2	3	1,937,568
128	0	106	11	96	6,453,968
128	1	0	128	8256	7,087,828
256	0	183	28	772	20,585,886
256	1	202	8	168	19,925,218

Fine DP cells and core visits are separate operation categories. Fewer payload cells do not imply less total work than ordinary Knuth at these sizes. In particular, the second 128-key input has global coarse ambiguity. No measured exponent or distributional bound on payload sizes is claimed.

Four ordinary mixed inputs at the same sizes reject all sixteen candidate scales and use Knuth. Three non-perfect uniform coarse inputs at $n = 32, 64, 128$ retain the entire payload table, as predicted by the explicit root-ambiguity example. Twelve further full queries reuse four perfect-size uniform preparations at $n = 31, 63, 127, 255$ and require neither fine DP cells nor new products. These uniform bases already admit direct linear construction. All 27 outputs agree with independent unrestricted Knuth and a separate original-weight tree traversal. The record `coarse_face_scale.json` retains every input, separate preparation and reference counters, all failed routes, exact accepted scales and error magnitudes, seed and eight source hashes. Bit-operation costs are separate from these arithmetic counts.

1.27 Reachable coarse-family grammar and rotation audit

The harness first compares the entire reachable interval and root-triple sets with literal optimal-tree enumeration on forty inputs through eight keys. It evaluates 10,275 literal tree costs. Twenty-seven separate perturbations select each uniform coarse optimum uniquely, checked against every competing small tree. Four exact scales verify the five-key strict rotation trap, its full optimum and all four neighboring costs. The three-key root-hole example is checked by an independent all-root recurrence.

For $B = 2, 4, 8, 16, 32$, a separate all-root computation builds the complete coarse-optimal grammar. Every interval in (188) is checked for reachability and for its entire optimal-root set. All 1,493 selected intervals also receive an explicit full-tree witness, whose original coarse cost equals the independent optimum. Selected counts are:

B	n	Selected intervals	Selected root triples
2	11	9	12
4	23	25	45
8	47	81	225
16	95	289	1377
32	191	1089	9537

For each size, a shuffled permutation of $1, \ldots, 2n+1$ gives distinct positive perturbations of all original key and gap weights. With $D = 2n \sum e_o$, the certificate holds. Independent Knuth construction and a direct full-tree cost traversal agree on all five fine inputs; each returned tree is also coarse-optimal. These are complete original-weight controls, not a new fast constructor.

The direct reference computations use 24,972 interval states and 1,349,726 root candidates. Their reachable graphs contain a total of 9,829 intervals and 47,070 root triples across the fixtures. Those are verification operations. Larger rows through $B = 2^{20}$ evaluate the proved formulas only, with no graph enumeration. The record `coarse_face_structure.json` retains the seed, all five original fine inputs, full costs, reference counts, exact scale controls and both executed-source hashes.

1.28 Adaptive coarse-grammar construction audit

The independent harness establishes the two-tree family before importing the new constructor. Direct all-root recurrences verify its complete reachable grammar at $m = 2, 4, 8, 16, 32$; literal tree enumeration checks the two smallest sizes. The persistent entry index is tested on 2,869 entries of separately generated condensed Monge matrices.

For 96 coarse inputs through twelve keys, including all-zero inputs, the prepared interval and root-triple sets equal those of an independent all-root recurrence. Fifty-six of these inputs also receive literal optimal-tree checks, evaluating 5,000 tree costs. Each preparation serves three independently perturbed rational fine inputs: all 288 full outputs match the independent original-weight optimum and remain coarse-optimal. These errors have both signs where nonnegativity permits. Ninety-six zero-budget preparations verify exact fallback, and the three-key root-hole fixture verifies that the missing middle root is excluded. Separate ordinary-input controls test automatic scale rejection. The direct reference computations total 20,900 states and 224,951 root candidates.

The two-tree scaling rows run complete generic preparation through $n = 383$. Each has the proved $V = n+1$, $A = n+2$, $Q = 4(n+1)/3$. Three full rational queries per preparation test generic refinement and the direct linear constructor, including the exact difference (96). Independent full Knuth controls cover the first six sizes, through $n = 191$; the largest row uses the proved two-tree classification and direct traversal of every original outcome. Additional perturbations deterministically select each of the two roots. All seven direct-constructor rejection controls fall back correctly.

n	Preparation core visits	Index nodes	Fine root candidates
5	6,834	47	7
11	40,522	187	13
23	174,784	632	25
47	602,090	1,827	49
95	1,893,664	4,948	97
191	5,557,466	12,965	193
383	15,476,304	32,590	385

Preparation includes both orientations, both unrestricted closure products and index construction. Core visits are one instrumentation category, not total instructions or a speed claim. Each generic fine query evaluates $n+1$ states and $n+2$ roots, checks $2n+1$ error terms and writes n keys. The direct constructor checks the same number of errors, sums $4m$ choice terms and writes the complete tree.

Three unstructured integer bases at $n = 32, 64, 128$ have respectively $V = 32, 65, 137$ states and $A = 32, 66, 143$ rules. Two rational fine queries per base match independent Knuth construction. Four additional full runs include automatic scale selection, generic preparation and output for the two-tree family. On the dense uniform control $n = 6B - 1$, the default budget finishes at $B = 2$ and exhausts at $B = 4, 8, 16$; every full output still agrees with Knuth. The retained record `coarse_grammar_audit.json` includes the seed, fine inputs, separate preparation/query counters, rejection work, verification scopes and eight executed-source hashes.

Combined-exchange control. For the nine-key example in Section 5.31, the 28 minimum-IPL shapes are checked against all 4,862 full shapes. The fourteen distinct coarse-preserving neighbors, combining every descendant promotion with one bottom-gap exchange, have error costs

$$224, 226, 227, 228, 229, 230, 230, 230, 231, 231, 234, 235, 238, 239.$$

The selected local tree has error cost 219, and the minimum among the 28 coarse optima is 212. At $D = 585$, full enumeration and Knuth give costs 34,734 and 34,727 for the candidate and optimum. Every descendant promotion, including those leaving the coarse family, is also checked directly. The source-hashed `coarse_exchange_control.json` retains the weights, candidate tree and every neighbor. This finite counterexample motivated optimizing the retained grammar directly.

1.29 Coarse-regret screening and root-range index audit

The harness first checks the signed allowance using independent all-root recurrences and minimum-height attaining trees, before importing the new constructor. It checks 540 optimal-subtree inclusions. Sixty generic finite Monge matrices of orders $n+1$, $1 \le n \le 15$, have independently generated signed boundaries and nonnegative densities. Dense minima verify 130,020 indexed root-range entries and 17,940 threshold enumerations. The hierarchy checks the per-depth $2K$ density charge, product-core bound, index-node bound, total inner dimensions and exactly $2n - 1$ range nodes.

The complete constructor audit checks 108 original-weight instances, 1,980 reached-interval root sets and, independently, all 1,980 threshold formulas using direct sums over interval outcomes. Each complete result matches an independent full optimum; 72 controls additionally evaluate 2,352 literal tree costs. Ninety-six of the 108 supplied coarse-base/scale pairs fail the earlier small-error certificate, yet screened refinement is exact. This statement concerns those supplied pairs, not every scale an automatic selector might try. Another 108 controls exhaust a zero expansion budget and return the independent Knuth optimum.

Automatic construction is checked on random mixed, near-uniform and geometric weights at $n = 16, 32, 64, 128, 256$. The first two families complete screening at each size; the geometric family exhausts the budget at each size and returns the exact fallback result. Four dense uniform-coarse-face controls complete at $n = 23$ and exhaust at $n = 47, 95, 191$. Five rational inputs, three unrelated inputs reusing one preparation, and twenty uniform or empty controls are also checked. Original-weight traversal verifies each returned full tree. The small independent recurrences evaluate 4,632 states and 15,500 root candidates.

A separate matched experiment uses positive random weights with up to 256 bits, at $n = 32, 64, 128, 256$, and compares four precisions on each identical input. Every row matches an independent full Knuth optimum. The record includes complete inputs, exact scales, original optimum costs, all failed work, and separate preparation and query counters. Table 2 reports selected counters; P is preparation core visits, Q is query range-minimum calls, and F is fallback root candidates. Dashes indicate that screened-grammar expansion was interrupted. These event categories are not a common unit of machine work, and V, A exclude preparation.

n	b	V	A	K	P	Q	F
32	1	–	–	116	172498	1685	1039
32	16	–	–	252	317835	2481	1039
32	36	127	187	262	326638	1183	0
32	216	33	34	263	328840	246	0
64	1	–	–	268	624306	4095	4034
64	16	–	–	653	1176520	5682	4034
64	49	–	–	666	1205482	7069	4034
64	343	104	112	679	1226437	988	0
128	1	–	–	654	2153460	9430	16062
128	16	–	–	1802	4749757	12460	16062
128	64	–	–	1880	4985710	17705	16062
128	512	301	333	1910	5056080	3349	0
256	1	–	–	1611	6335828	21299	66070
256	16	–	–	4805	16041939	27430	66070
256	81	–	–	5214	17327199	41114	66070
256	729	1029	1311	5269	17521921	14244	0

Table 2: Matched exact constructions for wide original weights. Precision $b = L^3$ completes screening at every tested size; smaller precisions often exhaust the counted budget. Preparation remains substantial.

At $n = 256, b = 729$, the screened grammar has 1,029 intervals and 1,311 roots, but preparation alone visits 17,521,921 core entries; the independent Knuth reference tests 66,070 root candidates. These counts establish neither a practical speedup nor an asymptotic exponent. The work bounds in Section 2.32 follow from the density-charge and regret proofs. All seventeen executed-source hash entries in the two records are checked against the retained sources. The records and reproduction commands accompany the manuscript; the theorem statements and proofs require no access to implementation files.

1.30 Implicit ordered references and nonlinear corridor audit

The interval-oracle harness establishes its claims with independent all-root tables before importing the optimized implementation. It checks 7,344 full root-cost sequences for the stated unimodality, 7,344 direct shape formulas, 756 literal interval-tree costs and 680 instances of (145). The optimized index then matches 17,424 interval values, 44,064 root-range minima and 44,064 complete threshold sets, including both orientations and partial valid-root ranges. Each threshold query also satisfies its binary-search call bound.

Full refinement checks 144 arbitrary original-weight queries, 54 literal full-tree controls and 144 zero-budget fallbacks. Two invalid coarse inputs are rejected, and three rational coarse inputs are reflected and reused for unrelated rational originals. Twenty-four larger full constructions at $n = 23, 47, 95, 191, 383, 767$ match independent Knuth. The six sparse-bump originals fail both

ordered exchange conditions, yet complete exact screening. For example, at $n = 767$ the coarse index stores 7,670 dyadic entries; the sparse-bump query reaches 959 intervals and 1,067 roots and makes 3,482 root-cost calls. The ordinary and densely perturbed failure cases retain all interrupted work before exact fallback. Preparation uses neither a Monge product nor a core list.

The nonlinear-corridor harness separately constructs literal expansion words before importing its solver. It verifies 1,638 exchange-regret inequalities and 530 optimal-prefix inclusions. The implemented class optimizer then matches 128 literal class optima and their state bounds. Full original construction passes 108 independent cubic controls, 84 literal full-tree controls, eighteen rational controls and twenty direct uniform controls. Forty-two scaling controls through $n = 1{,}535$ match independent Knuth: twenty-eight curved, irregular and reflected bounded-error cases satisfy the nonlinear-family guarantee; ordinary random and geometric cases exercise the unrestricted fallback.

Case	n	Retained states	Admitted transitions	Proved state cap
Curved, unit error	1535	26173	51320	53185
Curved, error two	1535	54437	107848	137205
Irregular increments	1535	17845	34664	33065
Reflected curve	1535	24384	47742	48635
Large curved	6143	183788	363478	372593
Large curved	24575	1007790	1999194	1965885
Large curved	98303	5295604	10525670	10026753

Table 3: Complete nonlinear-corridor constructions. Rows through $n = 1{,}535$ have independent Knuth controls. The three larger rows use the proved certificate and full original-cost traversal; no large Knuth run is asserted.

The larger curved instances use

$$f_i = i + \lfloor i^2/n \rfloor, \quad B = 2n \sum_i f_i + 20(n+1)^2,$$

with key error -1 at $i \equiv 0 \pmod 3$ and $+1$ elsewhere, and gap error -1 at odd positions and $+1$ at even positions. At $n = 98{,}303$, the fitted errors are $E_p = 32{,}767$ and $E_q = 98{,}304$, so $D = 622{,}576$. The forward affine fit from the earlier implementation has state upper bound 1,073,807,361 on the same input; this is a preflight bound, not an executed state count. The nonlinear corridor actually retains 5,295,604 states. The full Knuth table would contain 4,831,789,056 nonempty intervals, a formula-only comparison. These measurements check the implementation; the asymptotic guarantee is proved in Corollary 75.

A separate dense diagnostic covers eight residual patterns at seventeen sizes through $n = 512$, always with a baseline forcing minimum IPL on every interval. It rejects the proposed height-only final-core bound. The previous ruler theorem already gives the qualitative obstruction; the ramp observations motivate the exact maximum-support formula in Corollary 138. Dense diagnostic work and every failed route remain in the records. All three records include executed-source hashes and exact counters. The constructor and oracle bounds include fitting, preprocessing and reconstruction and exclude operand bit costs.

1.31 Range certificates and sharp box-threshold audit

The independent harness was written before the range-based constructor. It enumerates 6,917 trees through nine keys to check both shortening-capacity bounds, including 435 perfect-size gap

charges. It checks every joint key/gap box vertex at three and seven keys, as well as separate key-only and gap-only boxes: 33,304 exact all-root comparisons at the stated threshold. The explicit shifted-perfect construction is checked at fifteen sizes through $n = 65{,}535$, for three key/gap range pairs and below, at and above the threshold, giving 135 exact cost-difference identities. Fifty-four small boundary cases also have independent Knuth controls and test the full constructor's direct or fallback decision.

Full construction is compared with 576 independent cubic optima, including 96 rational cases checked by exact scaling to integer reference inputs, and 144 literal full-tree controls. Ninety-six fixtures separately satisfy the general range guarantee. Twenty-eight larger inputs through $n = 1{,}535$ are compared with independent Knuth. Fourteen curved or reflected inputs are certified by the new range condition although all fail the earlier $(n-1)E_0$ baseline gate. Ordinary random and geometric inputs retain all rejection work before exact fallback.

The larger curved family uses

$$f_i = i + \lfloor i^2/n \rfloor, \qquad B = n(f_{n-1} + 4) + 2,$$

with key errors -1 at multiples of three and $+1$ elsewhere, and gap errors -1 at odd indices and $+1$ at even indices. Three full constructions at $n = 6{,}143$, 24,575 and 98,303 are checked by the proved range certificate and direct original-weight traversal. These large rows have no quadratic-reference comparison. The corridor states and transitions are unchanged by subtracting the common baseline from the earlier curved family; all fitting, certificate and output counts are recorded separately. The certificate proof, rather than the measured state counts, supplies Corollary 79. The completed record retains exact inputs or formulas and eight executed-source hashes.

2 Adversarial core search and the odd-ramp proof audit

The search fixes nonnegative integer total mass at $W = n$ and maximizes

$$\frac{\max_{0 \le h \le H} \delta(E_h)}{n\sqrt{W\lceil \log_2(n+1) \rceil}}.$$

The core includes the n finite-completion diagonal entries. All layer costs and density counts use exact integers. A whole-table fixed point, when reached, justifies repeating the same count for subsequent layers. Single-unit and larger mass transfers, swaps, and reversals preserve the mass constraint. Simulated annealing affects only search acceptance; floating point is not used to compute costs or cores.

There are 48 starts over twelve sizes from 16 through 1,024, including non-power-of-two sizes, and 24 additional concentrated starts at six sizes. They perform 145,155 candidate evaluations in total. The first batch uses spread mixed weights, key-only weights, and uniform initial keys; the second uses spaced heavy keys, spaced ramps, clustered equal weights, and random sparse positions. Each batch first checks 232 complete layer vectors against the independent recurrence, with all-root and literal-tree checks on small inputs; selected winners receive further comparisons. Search budgets, seeds, full winning vectors, all layer counts, and operation counters are recorded. The table reports the best result at each displayed size across these starts.

$n = W$	Largest completed core	Normalized ratio
16	69	0.482152
32	185	0.417226
64	463	0.341792
128	1162	0.283691
256	2724	0.221680
512	6470	0.176604
1024	14582	0.134175

A separate diagnostic evaluates eight explicit patterns at fifteen sizes, for 120 complete layer vectors through 4,096 keys. These include a contiguous ramp, a contiguous random block, a spaced ramp, isolated key and gap mass, binary blocks, ruler-weight sampling, and random unit placements. The contiguous ramp gives 13,461, 36,276, and 98,168 completed core entries at 1,024, 2,048, and 4,096 keys, respectively. These measurements motivated the odd-ramp family with exact mass $W = n$ in Theorem 6. The proof uses the separately specified odd weights, not a fitted formula for the diagnostic's initial ramp.

The decreasing ratios in the first table alone would have been misleading as evidence for a universal near-linear bound. The actual conclusion follows from the theorem's explicit density coordinates. Search supplies candidate patterns and opportunities for falsification; it supplies neither an upper bound nor an asymptotic tightness certificate.

Independent proof controls. The new mathematical audit checks 2,652 straight and 2,652 bent rotation identities, 894 dyadic partitions through $n = 129$, and 4,277 scale bounds. The odd-ramp audit compares its subsequence recurrence with 445 literal subset enumerations and checks 388 perfect-slot embeddings against an all-root interval recurrence. There are 42,748 independent checks of the rightmost-zero formula and 17,688 checks of the closed-form density against subsequence DP. At $n = 1024$, all 1,896 cells in the proved set also match the original penalty-layer recurrence. That original-layer audit and the smaller controls evaluate 4,736,120 dense states and 8,852,066 root candidates. They are diagnostics, not operations of the sparse construction algorithm.

$n = W$	Proved positive cells	Subsequence DP	Original-layer DP
1024	1896	checked	checked
4096	15792	checked	—
16384	128736	formula only	—
65536	1039296	formula only	—

The last two rows evaluate the proved coordinate count and densities; they are not additional independent full-layer computations. The lower bound holds for every $n = 2^{2t}$, $t \geq 5$, by its proof. These records are in `core_size_search*.json`, `core_size_patterns.json`, `core_size_lemmas.json`, and `ramp_core_tightness.json`. Each records hashes of the exact sources used.

3 Local tile charging: probe and exact one-key audit

The tile-mass probe evaluates mixed random, sparse, uniform, geometric, and concentrated inputs, followed by fixed-mass local mutations. Its objective is the ratio of a maximal tile's density mass to the mass of its own span. The 1199 evaluations include 123 independent all-root controls. They examine 19,699,184 dense states, 38,539,843 root candidates, and 1,169,332 tiles. The largest

observed ratio is four at $n = 256$, $h = 7$, with one key carrying all the weight. Scaling that weight to one leads to Theorem 142; the search alone does not establish the unbounded ratio.

The separate formula audit checks every interval value and every interior density of the explicit family through $n = 1024$. It uses all-root recurrences through $n = 64$ and monotone-root recurrences at larger sizes. There are 700,024 interval-formula checks, 697,992 density-formula checks, and 1023 penalty-line checks. The exact completed core sizes are:

n	h	Completed core	Tile support	Tile mass
16	3	33	0	0
32	4	71	1	1
64	5	149	2	2
128	6	307	3	3
256	7	625	4	4
512	8	1263	5	5
1024	9	2541	6	6

The general one-key corollary is additionally checked on 1316 cases: every key position through $n = 16$, two positive masses, and all tested layers, using literal all-root recurrence. These contribute 88,170 additional density checks. Together the formula harness records 6,387,678 dense diagnostic states and 12,975,174 root candidates. Larger rows through $h = 30$ evaluate only the proved formulas and are explicitly labeled as such. They are not independent dense solves.

The executed sources and exact operation records are retained in `dyadic_mass_probe.json` and `single_key_tile.json`. The proofs in Section 5.9 establish the asymptotic local obstruction and global linear bound independently of these finite checks.

4 Independent mixed-slot construction audit

The reconstruction procedure is checked on all 32,902 nonempty slot subsets of perfect trees through height four. Every surviving key has comparison length no larger than its selected slot's length, and every gap satisfies the neighboring-key identity.

The constructor has 717 independently checked complete solves, including 473 literal slot-minimum and literal tree-minimum controls. Its independent height recurrence examines 558,638 interval states and 15,547,880 root candidates, including the smaller scale controls. The positive four-key example in Section 2.8 has full optimum 121; literal minimum-internal-path-length enumeration gives 218. The small constructor runs record 82,568 slot states, 145,171 suffix-level updates, and 69,863 lower-level scans. Preprocessing, traceback, array initialization, node deletion, and output have separate counters.

The complementary above-capacity formula is checked on 23,920 intervals against full mixed penalty layers, and 18,000 ordinary densities agree with their zero-gap counterparts. Those independent layer computations examine 184,056 states and 1,300,792 root candidates.

There are 53 scale rows through perfect capacity 4095, using positive mixed weights with baselines 2^{60} on keys and 2^{61} on gaps. Twenty-three smaller rows also use the independent all-root recurrence; the larger rows check reconstructed height, direct original cost, and the slot objective, with their independent-reference flag set to false. At height twelve the operation counts are:

n	r	Slot states	Suffix updates	Lower-level scans
4095	0	4095	8177	4083
4094	1	8188	16351	8165
4093	2	12279	24522	12246
4091	4	20455	40852	20402
4087	8	36783	73468	36694
4079	16	69343	138516	69190

Each row tabulates 4095 valuations, builds 4095 perfect-tree nodes, deletes exactly r nodes, and outputs all n original keys. The stated asymptotic bound follows from the valuation-sum proof, independently of these rows. The executed source hashes and operation records are in `mixed_slots.json`.

5 Binary-weight search and exact gap-core audit

The binary diagnostic performs 84 searches: seven power-of-two sizes, three placements of positive weights, and four initial patterns (random, alternating, centered, and prefix). Mutations exchange a zero and a one; occasional reversals preserve that invariant. Mixed runs have mass n; key-only and gap-only runs have mass $n/2$. The objective is the maximum completed core across all terminal-penalty layers, including the completion diagonal. Whole-table fixed points allow identical remaining layers to be reported without recomputation.

The search evaluates 117,084 candidates, recording 40,476,345,560 dense states, 80,519,888,452 root candidates, and 40,301,399,592 density evaluations. These are diagnostic operations, not sparse-constructor costs. An independent Python reference checks 81 vectors, including every winning vector through $n = 128$; its small cases include all-root and exhaustive-shape controls. The largest value among the four starts in each placement is:

n	mixed	keys only	gaps only	proved gap upper bound
16	69	50	44	44
32	182	137	100	100
64	465	341	228	228
128	1151	861	516	516
256	2762	2118	1156	1156
512	6473	5032	2564	2564
1024	14701	11571	5635	5636

The last column follows from Theorem 12; it is an upper bound over every fixed-mass binary gap input and all layers. The mixed and key-only columns support no asymptotic assertion.

A separate all-root recurrence checks 2,189 inputs: all binary gap vectors through eight keys, all vectors over $\{0, 1, 3\}$ through five keys, seeded nonbinary vectors, four search winners, and four explicit alternating family members. It checks 5,282 mass identities, 32,899 one-missing-slot formulas, 22,335 capacity densities, 157,129 vanishing above-capacity densities, and 3,976 whole-triangle identities. The explicit family contributes 10,524 independently evaluated density formulas. Total reference work is 492,579 states and 8,892,512 root candidates. The binary bound is checked on 60 eligible layers. Exact weights, seeds, operation counts, and source hashes are retained in `binary_core_search.json` and `gap_core_identity.json`.

6 Mixed boundary-mass audit

An independent all-root recurrence checks 308 inputs: every mixed binary vector through three keys, seeded integer vectors through sixteen keys, and alternating families through $n = 128$ with four baseline pairs. The audit checks 756 mixed mass identities, 6,656 one-missing-slot formulas, 3,376 densities immediately below capacity, 136,680 above-capacity densities, and 80 closed family mass formulas. Reference work is 369,396 states and 13,048,220 root candidates. The source hashes and exact family rows are in `mixed_boundary_mass.json`. These checks accompany the proofs in Section 5.10; the asymptotic conclusions follow from the displayed identities.

7 Capacity-band construction audit

The independent all-root reference covers 272 mixed inputs: exhaustive binary weights through three keys, seeded integer vectors, and geometric weights through 32 keys. Across widths $0, 1, 2, 4, 8, n$ where distinct, it checks 253,303 retained values, 1,287 complete reconstructed trees, and 4,100 per-layer candidate bounds. Reference work is 94,179 states and 924,277 root candidates. The band solver uses 199,627 states and 366,637 root candidates, with initialization recorded separately.

Sixteen scale rows use near-uniform strictly positive key and gap weights, heights six through twelve, and widths $0, 1, 4, 16$. Each full instance has $n = 2^h - 1 - d$, so positive widths include nonperfect sizes. These larger rows verify reconstructed original-weight cost and operation counters; their independent-reference flags are false. At height twelve:

n	d	base entries	band states	root candidates
4095	0	4096	40974	40974
4094	1	8189	118792	196622
4091	4	20450	343803	702361
4079	16	69224	1174685	2621548

Each row additionally scans n input positions and outputs all n keys. These counts cover all retained intervals across all twelve layers, rather than one slot-DP instance. The asymptotic bound follows from band closure and telescoping root ranges in Theorem 13. Exact rows and executed-source hashes are in `capacity_band.json`.

8 Capacity interaction audit

The independent all-root audit includes every mixed binary vector through five keys, seeded integer inputs through 32 keys, and unit-input controls. It checks 28,259 capacity identities, 6,416 telescoping totals, 5,856 subadditivity bounds, and 5,856 width-one interaction inequalities. Its 3,091 vectors use 371,731 states and 3,302,476 candidate roots. It also retains the four-key wider-band counterexample from Section 5.11.

The separate sharp-family audit checks heights two through six, including 28 independently solved unit inputs at the smaller sizes. It verifies five alternating-family formulas, three direct interaction differences, and five uniform-key wider-tail formulas, using 194,152 states and 6,131,120 root candidates. Larger rows through height 128 are closed-form evaluations only. These records are `capacity_interaction.json` and `capacity_interaction_sharp.json`.

An initial optimized dense diagnostic investigated the proposed all-width allowance, maximizing $[T_d(w) - \sum_x w_x T_d(e_x)]/(hW)$ over every band width and early layer. It uses 15 fixed-mass searches

through 256 keys, with 1,200 mutations each and three starting patterns per size. Structured inputs include uniform keys, uniform gaps, mixed weights, alternating keys, odd ramps, and seeded random weights. Independent all-root controls check 150 optimized vectors. Including unit-profile construction, structured evaluation uses 157,388,626 states and 313,030,555 root candidates; search uses 1,667,827,872 states and 3,310,832,256 candidates. The largest observed ratio is 1223/1536, attained by uniform keys. The sparse family in Theorem 146 subsequently disproves this allowance. Exact weights, seeds, counters, and executed-source hashes are in `core_interaction_search.json`.

9 Sparse interaction falsification and formula audit

The one-sided all-root diagnostic checks 2,956 vectors and 91,370 inequalities, using 1,770,275 states and 28,876,151 root candidates. It finds the ten-key above-capacity counterexample in Section 5.11.

The subsequent sparse diagnostic evaluates 19,614 inputs, including every unit pair on keys and gaps through 64 keys, structured unequal-weight pairs, and sparse random inputs through 512 keys. Its optimized recurrence includes unit-profile construction: 4,587,178,568 states, 9,062,247,119 root candidates, and 19,197,336 band evaluations. Independent all-root controls verify the four retained winners through 64 keys. The observed ratios exceed one and grow with input size; these finite violations motivated the explicit family of Theorem 146.

The family audit checks the combined input and both unit inputs against independent all-root recurrences through height seven. It verifies 123,585 eligible density cells, 920 band entries, and three complete family identities, using 870,624 states and 66,536,256 root candidates. Literal rank choices verify 32,424 local table entries through height twelve, with 415,042 feasible rank pairs counted. Larger rows through height 128 evaluate only the proved closed forms. These are separately labeled in `sparse_interaction_barrier.json`. The other records are `one_sided_interaction.json` and `sparse_interaction_search.json`; all retain executed-source hashes.

10 Rank-reward stability and expanded tail diagnostics

The rank-template audit checks 432 small inputs. Canonical maximizing sequences are independently compared with 27,717 literal subsets over 3777 intervals. It verifies 6817 coordinate monotonicity inequalities and 15,930 single-rank perturbation bounds. Ruler clipping is checked on 75 larger fixtures through 256 keys, giving 45,840 inequalities; 540 upper densities are compared with independent all-root mixed layers. The generic subsequence work totals 286,337,285 matching candidates and 1,350,475 stored interval scores. The record is `rank_reward_tail.json`.

The periodic obstruction has a separate audit through $h = 7$. It checks 491 boundary entries, 20 family identities, and 11,036 upper density cells using 2,429,031 subsequence candidates. At $h = 3$, 48,620 literal subsets independently check all 55 eligible interval scores. Larger rows through $h = 128$ evaluate only the proved closed form. These modes are labeled in `periodic_tail_barrier.json`.

The original total-tail target $T_{h,d} = O(W(L + c_h/d))$ is tested separately. A sparse diagnostic evaluates 23,183 inputs through 1024 keys, including all unit pairs on keys and gaps through 64 keys, unequal pairs, phase-changing combs, ramps, and random small supports. Its largest exact ratio to $W(L + c_h/d)$ is 485/168. The optimized recurrence records 17,157,958,100 states, 33,808,959,382 root candidates, and 47,383,400 band evaluations. Five retained winners through 128 keys have independent all-root controls. A second diagnostic evaluates 462 repeated sparse clusters through 2048 keys; its maximum is 2828/1107, with 3,107,692,416 states, 5,649,382,012 root candidates,

and 3,997,440 band evaluations. Its two small retained winners also have independent all-root controls. These finite maxima do not prove an upper bound. The records `sparse_tail_search.json` and `cluster_tail_search.json` retain exact weights, objectives, operation counts, and executed-source hashes.

11 Alternating-tail matching and sharp-scale audit

The matching audit compares generic rank-subsequence scores with an independent cardinality-matching recurrence on 2887 inputs. It checks 147,191 endpoint inequalities and constructs 41,517 augmenting paths in 18,703 packings, verifying every resulting matching, removed weight, and repair inequality. There are 58,142 global tail-bound checks. The inputs include all binary vectors through nine keys at every admissible odd capacity, and random integer weights through 128 keys. Sixteen cluster instances through $C = 128$ check (25), 1045 endpoint formulas, and 32 terminal scores. Eleven spaced-maximum families check the additive scale. Larger cluster rows through $h = 128$ use only the proved formula. The record is `alternating_tail_bound.json`.

A separate compiled probe tests the wider clipped-ruler candidate $U_d/[P(t + (c + 1)/d)]$. It evaluates 25,683 inputs and records 84,957,303 interval scores, 4,737,828,651 candidate transitions, and 82,044,166 upper densities. All 451 small control vectors agree with the generic subsequence recurrence. Its maximum observed ratio is 321/292, on a height-seven template with two unit keys. This disproves coefficient one in that inequality but supplies no asymptotic upper estimate. The exact weights, seed, counts, and source hashes are retained in `periodic_tail_probe.json`.

12 Rank-weighted repair certificates and exact barriers

The paired canonical diagnostics each evaluate 7225 inputs, including all ternary interior vectors on ten positions and structured, random, and sparse inputs through 64 positions. Each run has 6976 independent generic-subsequence controls, 224,622 endpoint checks, and 26,402 global certificate checks. The leftmost run explicitly constructs 1,385,304 repair paths and finds 13,185 losses strictly above the true loss; the rightmost run constructs 546,117 paths and finds 7087 strict losses. Every path is checked for feasibility, cardinality, and its full signed retained-rank accounting. The records are `rank_matching_repair.json` and `rightmost_rank_repair.json`.

The separate exact-family audit checks 3786 zero-loss equivalences on 306 inputs, and 16 instances of Lemma 149, including weights above 2^{64}, against 160 literally enumerated old matchings and independent shortened scores. For Theorem 150, heights three through ten give 4088 explicit rightmost boundary witnesses and repair-loss formulas, 225 independent scalar-DP witness controls, and eight complete certificate formulas. Every optimal starting matching is enumerated at both height-three thresholds and the largest height-four threshold, giving 1023 checks of tie-independent loss; the two height-three optimum sets are also checked by literal enumeration without a forced-prefix assumption. There are 32,352 explicitly checked repair paths in this family audit.

Full generic subsequence tables independently give zero true tails through height five. Eight explicit constructions check the exact component counts of Corollary 151. Larger rows at heights 16, 32, 64, and 128 evaluate only the proved formulas. The independent scalar controls use 41,799,787 transitions; the generic controls use 225,793 candidate transitions and 6637 scores. Exact inputs, operation counts, and executed-source hashes are retained in `rank_repair_barrier.json`. The finite runs audit the proofs; the unbounded ratios follow from the displayed families.

13 Weighted cluster tails and input sensitivity diagnostics

For integer weights and an integer template in $[0, t]$, a right input-key derivative is computed exactly by maximizing the pair (ordinary score, reward at the perturbed key) lexicographically in each interval. An independent evaluation uses

$$D_x U_d(p) = U_d((t+1)p + e_x) - (t+1)U_d(p).$$

A nonzero primary-score gap is at least one; scaling by $t+1$ prevents any slope reward, at most t, from closing it. Among primary maximizers, the perturbation selects the largest slope. Taking the signed tail sum preserves the identity. No finite-difference rounding is involved.

The weighted-cluster audit compares generic subsequence and independent cardinality-matching tables on 81 inputs, including all ternary vectors on a three-key cluster, unequal weights, zero minima, and ties. It checks 3652 endpoint formulas, 162 terminal scores, 169 coordinate derivatives, and 47 joint increments. The generic recurrence uses 44,724,022 candidate transitions and 308,926 scores; the matching recurrence uses 28,460,181 transitions and 187,123 scores. Larger formula-only rows are explicitly labeled. The record is `weighted_cluster_tail.json`.

A separate full-ruler diagnostic checks 21,020 fixed-mass inputs through 256 positions and records 3,119,522,724 candidate transitions, 20,512,788 scores, and 19,108,544 density derivatives. All 29 independent scaled perturbation controls agree with its lexicographic score/slope recurrence. The maximum derivative ratio to $h - 1 + 2^h/d$ is 416/129, on 128 positions. This finite value implies no asymptotic bound or obstruction for the full ruler. A preliminary canonical-witness probe checks 5616 Euler identities on 608 inputs and computes 11 exact directional controls. At ties its canonical coefficients can differ from right derivatives; these quantities remain distinct in the records `key_sensitivity_search.json` and `key_tail_sensitivity.json`. All records retain exact inputs, operation counts, seeds, and executed-source hashes.

14 Threshold falsification and boundary-tile diagnostics

The nested-threshold probe evaluates 14,432 inputs and 35,722 width comparisons, including exhaustive ternary interior weights on ten positions. It records 1852 upper-relation violations and 4003 reverse-relation violations. The counterexample in Lemma 153 is separately checked by literal enumeration of 876 subsequences for 36 interval scores, comparison with the generic subsequence recurrence, and three original all-root terminal-layer controls. The records are `nested_tail_probe.json` and `nested_tail_example.json`.

The broad boundary-plus-mass diagnostic evaluates 21,436 inputs, with structured inputs through 1024 keys, non-power-of-two sizes, sparse unit outcomes, and fixed-mass mutations over keys, gaps, and mixed outcomes. It scans 87,706,185 accepted tiles. Its largest observed ratio $K(T)/(s+W(J_T))$ is 121/136, on a gap-only input. All 93 small independent all-root controls agree with its layerwise maxima; these controls enumerate 15,301 tiles independently. The compiled run records 3,240,243,236 states, 6,584,091,125 root candidates, and 3,217,996,956 density evaluations. The record is `tile_boundary_probe.json`.

The maximum-over-all-tiles objective often settles on a small tile. A second search therefore restricts its objective to capacity-adjacent tiles with $s \geq 4$ and $c \geq 4s$, preserving total mass $n/4$ or n. It evaluates 37,224 inputs through 512 keys and visits 212,592,240 tiles; the largest eligible ratio is 11/14. Six retained small winner tiles agree with independent all-root layers. It records 8,185,066,752 states, 16,269,275,134 root candidates, and 8,131,873,536 density evaluations. The record is `tile_capacity_support.json`. Both searches are finite falsification diagnostics; neither proves (191).

The one-key square audit checks 1,038,774 maximal squares over all key positions at selected sizes through 128 keys, including non-power-of-two inputs. It also checks 1300 arbitrary squares, including non-dyadic sides, and compares 155 full support sets against independent all-root layers. The proof of Lemma 178 uses only the ruler-pair classification and interval geometry. The record is `one_key_square.json`. All records retain exact inputs, operation counts, seeds where used, and executed-source hashes.

15 Full-ruler clipping sharpness and directed support searches

The audit of Theorem 18 enumerates feasible ranks of the two positive keys on 16 parameter pairs, through 1024 keys. It checks 3544 compatible score decompositions, 3544 coarse score mappings, and 16 parity sums. Twenty generic subsequence controls independently check full and clipped tails through 128 keys, using 2,370,068 score transitions and 20,424 densities. Twelve original all-root controls through 64 keys include nonzero gap weights and a key-weight scaling factor of seven; they use 1,473,440 root candidates and 71,664 states. Literal enumeration checks 11,520 subsequences for 18 boundary scores.

The sparse-rank boundary counts in Lemma 154 have 28 independent two-key rank controls against the generic subsequence diagnostic, plus 840 literal subsequences for 14 scores on the smallest instance. Across both families, the rank audit uses 174,756 one-rank candidates and 433,864 two-rank candidates for 14,276 scores. Larger formula-only rows are marked separately. The records are `ruler_clipping_sharp.json` and `spaced_rank_tail.json`.

A preceding joint input/template search evaluates 22,212 inputs through 128 positions. Its largest observed ratio to $P(\max v + \sum v/d)$ is 16/17. The explicit family in Lemma 154 instead has unbounded ratio. The search records 865,917,888 score transitions and 9,089,940 densities; three retained small tails have 34,560 literal-subsequence checks for 54 scores. The record is `template_tail_probe.json`.

A separate fixed-side, variable-integer-mass search attacks (191). It evaluates 40,557 inputs through 256 keys and visits 121,969,824 accepted tiles. Its maximum observed ratio is 136/149, at side 32 with 136 positive entries and local mass 117. Fifteen retained small winner tiles agree with independent all-root layers, using 2,266,528 candidates and 109,936 states. The compiled search records 2,806,155,040 states, 5,480,712,131 root candidates, and 2,774,094,048 density evaluations. These finite tests leave the local support hypothesis open. The record is `tile_support_frontier.json`. All records retain exact inputs, operation counts, and executed-source hashes; seeded searches also retain their seeds.

16 Unit-support geometry and off-envelope interactions

The single-gap density formula has 274,066 independent ordinary-density controls, using 3,079,878 all-root candidates and 307,625 states. The geometric audit checks 78,240 maximal dyadic tiles, 688 literal envelope counts, 248 single-key support sets, and 1222 arbitrary squares against the weaker $8s$ bound. The largest observed envelope-to-side ratio is 511/256, consistent with the proved $2s-1$ family. The record is `unit_support_envelope.json`.

A directed two-outcome search excludes every envelope cell from its support objective. It evaluates 819 inputs through 2048 keys and scans 22,696,956 tiles. Its counts are 3,760,248,528 states, 7,001,021,069 root candidates, and 3,755,113,776 density evaluations. Six retained small winner tiles agree with independent all-root layers, using 718,432 candidates and 35,984 states.

This search suggested the explicit family; the asymptotic obstruction follows from Theorem 155. The record is `interaction_support_probe.json`.

The explicit-family audit checks $s = 8, 16, \ldots, 1024$, hence $n = 64, 128, \ldots, 8192$. It verifies 36 density-one witnesses and 144 four-corner score formulas by independent feasible-rank enumeration, using 23,808 single-rank and 2,090,940 pair-rank candidates. At $n \leq 128$, twelve scores also agree with a generic subsequence recurrence, using 302,446 transitions and 2806 score entries; three densities agree with original all-root layers, using 2,375,360 root candidates and 59,936 states. Larger cases use feasible-rank enumeration, rather than a dense all-root computation. The record is `unit_envelope_obstruction.json`. All three records retain executed-source SHA-256 manifests. Verification counts do not enter the proofs of the gap formula, geometric envelope, or interaction family.

17 Pair amplification and exact binary dilation

The fixed-tile amplification search evaluates 2470 inputs through 1024 keys, including translated pairs, shared endpoints, unequal weights, and local integer-weight mutations. It records 1,858,339,360 states, 3,315,818,765 root candidates, and 6,790,720 density evaluations. Six retained tiles have independent all-root controls using 7,126,080 candidates and 179,808 states. The largest observed ratio $K(T)/(s + W(J_T))$ is $12/25$; no upper bound follows from this finite maximum. The record is `pair_amplification_probe.json`.

The independent binary-dilation audit checks saturation on 1536 cases using 21,588 literal subsequences, with arbitrary nonnegative rank rewards. The two scope counterexamples use twelve additional literal subsequences. Forty dilation instances give 6142 score controls and 4392 density controls from separate generic subsequence recurrences, using 416,654 transitions and 8958 score entries. Original all-root layers with arbitrary nonzero gap weights check 2286 entries and 1434 densities, using 398,656 root candidates and 38,304 states. All 42 positive controlled fine densities are outside the unit-outcome envelope. Rounded endpoints inside zero blocks are included. The record is `binary_dilation_audit.json`.

A larger original-layer audit checks 27 replicated pair, alternating, and three-block patterns through 1024 keys. Every instance preserves the coarse tile's core cardinality and density mass; for $d > 1$, every positive fine density is off-envelope. This run uses 22,874,208 states, 40,616,815 root candidates, and 84,672 density evaluations. It uses monotone roots; the preceding audit supplies independent all-root controls at smaller sizes. The record is `binary_dilation_family.json`. All three records retain executed-source SHA-256 manifests. The exact identity and support transport follow from Section 5.19, independently of these finite checks.

18 Plateau cores and rank-threshold compatibility

The endpoint-cut search evaluates 1960 inputs through 1024 keys, including replicated coarse patterns, unequal blocks, positive endpoint fringes, and integer-weight mutations. It records 1,707,458,304 states, 3,241,769,152 root candidates, and 6,204,928 density evaluations. Fourteen retained tiles have independent all-root controls using 16,627,520 candidates and 419,552 states. The largest observed ratio $K(T)/(s + W(J_T))$ is $20/37$. This finite maximum proves no upper bound. The record is `endpoint_amplification_probe.json`.

The one-block audit checks 5561 complete above-capacity support sets against the original all-root recurrence, including nonzero gap weights and positive endpoint cuts, through 64 keys. Its 1,143,360 density controls include 635,883 positive-endpoint zero identities. The original recurrence

uses 33,319,148 root candidates and 2,723,869 states. Independent literal visibility enumeration checks 193,533 pairs; 23,375 block rewards agree with both direct ruler sums and (26). Constructed support sets include the exact values, not only cardinalities. The local square corollary has 50,032 controls, exhaustive through twelve keys and sampled at larger sizes. Twelve larger sharp-family constructions reach 32,768 keys; these use the linear constructor rather than the original recurrence. At the largest size it computes 16,383 ruler values, performs 16,382 sliding updates, pushes 16,383 and pops 16,369 stack indices, considers 32,738 pairs, and outputs 16,356 densities. The record is `plateau_core_audit.json`.

The complementary below-capacity construction. The independent audit covers 1025 block inputs: every contiguous block through 17 keys, selected cases through 64 keys, rational common weights, and heights through twice the binary length of n. It compares 8812 complete cores with the original all-root recurrence, checking 1,070,212 ordinary densities, 88,033 top-sum interval values, and 222,434 local tiles. The top-sum identity also has 336 arbitrary-template controls with literal subset enumeration. The original references use 18,032,986 root candidates and 1,556,996 states. Sixteen larger sparse constructions reach 65,536 keys and enumerate every layer through twice the binary length. Their checks use the explicit coordinate identities, without a large original all-root reference. Counts and final source hashes are in `plateau_below_audit.json`.

The separate fixed-tile search maximizes $K(T) - W(J_T)$, using variable integer mass with key-only, gap-only, and mixed mutations on both sides of capacity. It evaluates 34,692 inputs through 640 keys, using 22,971,840 tile-density checks and 11,439,157,710 optimized root candidates. Twenty-seven small winners agree with independent all-root layers, using 9,000,132 candidates. The maximum observed normalized excess is $39/64$, from $K = 63$, $W = 24$, $s = 64$. The record `tile_excess_probe.json` retains full winners, counts, seed, and source hashes. These finite results do not prove the proposed $K = O(s + W)$ bound; Theorem 26 has an independent proof.

The rank-threshold discovery probe evaluates 65,454 inputs using 15,176,940 subsequence transitions. Its first obstruction has 22 keys, full reward seven, and threshold optima $(5, 2, 1)$. Independent literal enumeration checks all 170,544 length-15 subsequences of that input. The explicit family in Theorem 158 and the smaller example are checked by all 32,752 single deletions through 16,384 keys, evaluating 425,838 retained positive ranks and 5,176,333 threshold comparisons. For $n \leq 128$, 22 independent generic subsequence controls use 145,386 transitions and 66 score entries; four original all-root controls with nonzero gaps use 2,812,064 root candidates and 73,456 states. The larger rows enumerate every deletion and are not original all-root runs. The record is `rank_threshold_obstruction.json`, which also hashes the raw search artifact. All three audit records retain executed-source SHA-256 manifests. The theorems follow from their proofs independently of these finite checks.

19 Finite band products and complete two-block cores

The generic band audit checks 1332 inputs, including exhaustive binary sequences through $N = 4$ and signed integer sequences through $N = 16$. It compares 28,024 finite product entries against 2,098,072 literal pair choices, including infeasible completion entries, and checks 23,821 feasible rewards. A further 3996 coarsened-core and 11,808 coarsened-product entry controls test the local aggregation in Lemma 46. These checks test exact values and the finite penalties, not only the support constants.

The interior audit checks 576 support sets on 168 mixed inputs through 32 keys against original all-root layers, with 5690 densities and 672 literal subsequence intervals from 20,517 choices. Its original recurrence uses 1,345,248 root candidates and 146,592 states. Twenty-four larger condensed constructions reach 16,384 keys, with 661 additional square controls. Those rows are constructor checks rather than original all-root runs. The record is `band_dominance_audit.json`.

The complete audit checks 952 mixed inputs through 64 keys, including all 792 binary words with exactly two positive runs through ten keys and 160 seeded larger inputs with common positive key weight three. Nonzero gap weights are included. All 2953 complete above-capacity support sets and 334,032 densities agree with the independent all-root recurrence, which uses 12,507,215 root candidates and 784,506 states. Fourteen larger full constructions reach 8192 keys and include 864 local square controls. For two unit keys at that size, the constructor evaluates 31,017 distinct density candidates and outputs 3219 positive densities. It performs 26,527 range queries and 15,631 condensed-matrix value queries; the record separately counts the reused sparse-product operations. These larger rows do not have original all-root references. The record is `two_block_core_audit.json`. Both records retain executed-source SHA-256 manifests. The bounds follow from the finite product, visibility, and endpoint-case proofs, independently of the finite audits.

20 Finite rank transfers and dense-core audit

The audit exhausts 363 inputs over key weights $\{0, 1, 3\}$ through five keys. It compares 23,232 finite transfer entries, including infeasible completion entries, with independent dense products using 459,456 candidate additions. All 11,385 feasible entries also agree with a separate subsequence DP and 52,245 literal subsequences. Thirty-two rational-weight inputs add 1048 exact entry controls.

For 70 mixed inputs through 32 keys, including geometric, increasing, near-uniform, constant, and random key weights with nonzero gap weights, 1172 original all-root layer scores agree with (160). Separately, 32 many-block inputs with varied positive-block lengths have 128 original-layer support-set comparisons and 3544 interior density controls. Together these original all-root controls use 987,840 root candidates and 104,460 states.

One hundred chains of signed unary functions supply 3850 feasible-score controls against an independent state DP. The explicit width-one family in Theorem 163 has 1493 formula entries checked against both that DP and condensed products, and 341 density-one square cells. Six weighted-ramp constructions through $c = 255$ check 33,138 transfer formula entries and 32,136 densities from Theorem 161. The four smaller rows, through $c = 63$, also have 2130 independent subsequence-DP entry controls. The two larger rows use the condensed product and the closed form, without that additional DP control. At $c = 255$, $r = 128$, the strict feasible strip has 12,097 positive entries and the side-64 square has 2048. The product has 127 chain multiplications, whose total input-core count is 206,530; its internal multiplication operations are recorded separately.

The record is `rank_transfer_audit.json`. It preserves exact operation counts and executed-source SHA-256 manifests. These finite checks verify the implementations and identities at the recorded sizes; the dense families and construction bounds follow from the proofs.

Slot-window products and full prescribed-height trees. The independent slot-window audit exhausts 363 inputs over $\{0, 1, 3\}$ through five keys. All 23,232 finite product entries, including infeasible completion entries, agree with separately evaluated dense products using 459,456 candidate additions. The 2946 feasible entries also agree with an independent weighted-subsequence DP and 20,340 literal subsequences. Twenty-five rational single-run cases, including runs longer than capacity, check another 6820 finite factor entries.

One hundred mixed inputs through 24 keys use constant internal gaps, arbitrary endpoint gaps, and uniform, geometric, zero, or random keys. The complete constructor agrees with 200 original all-root height optima; every output tree has independently checked inorder, height, and original mixed cost. The references use 235,160 root candidates and 34,270 states. The audit also checks 220 insufficient-capacity rejections and the explicit rejection of unequal internal gaps.

Nine checkerboard squares through side 512 verify 349,524 four-corner transition densities against (158). Eight larger uniform or three-run constructions through 8192 keys include the sparse product, witness traceback, full tree repair, and independent output-tree cost and height. Uniform rows additionally match the closed minimum-IPL optimum; the larger three-run rows have no independent dense optimum reference. At 8192 keys and height 14, the three-run construction has 61,431 input core entries, two run-product multiplications, and a final core of 20,147 entries. Leaf traceback scans the 16,383 slots twice in total, once for buckets and once for output. The record `slot_run_audit.json` contains exact operation counts and seven executed-source hashes. The theoretical bounds and signed obstruction follow from the proofs above.

Vector construction with arbitrary mixed gaps. The subsequent audit exhausts all 680 mixed binary inputs through four keys. Literal enumeration checks 7880 tree-cost normalization

identities. Independent prefix enumeration checks 12,496 state values and their attaining predecessors using 53,824 selected-slot sequences. Forty-eight finite forced-end factors supply 15,600 entry comparisons with the prior condensed window factor and 14,256 four-corner Monge controls, including infeasible entries and rational weights.

Another 84 mixed inputs through 32 keys include geometric weights, unequal gaps, rational weights, long key runs and isolated gap changes. Together the two input groups give 1528 comparisons with the original all-root height optimum, each with independent output-tree order, height and original mixed cost, plus 1184 insufficient-capacity checks. The reference recurrence uses 828,560 root candidates and 107,736 states.

Twelve larger complete constructions through 8192 keys cover uniform keys, three key runs, and three key runs with exceptional internal and endpoint gaps. Uniform rows additionally match the closed minimum-IPL optimum; the other larger rows have output-tree checks but no independent large optimum oracle. At 8192 keys and height 14, the case with three exceptional internal gaps has five transformed key runs, six long-run passes and five singleton passes. It makes 752,788 finite run-entry queries, processes 61,437 singleton states, and stores 180,224 predecessor slots. The record `slot_vector_audit.json` retains the separate exact counters and eleven executed-source hashes. SMAWK is the published primitive cited in Theorem 30; the full construction bound follows from its finite Monge interface and the valuation-sum amortization.

Sufficient unrestricted height and full vector construction. The height audit exhausts 22,140 mixed inputs over $\{0, 1, 9\}$ through four keys. A further 72 inputs through eight keys bring the literal shape count to 323,604, including 30,955 optimal shapes. The audit computes μ and z independently from every interval. It checks 152,072 sorted marked-set cuts and 790,245 positive-subtree depth assertions across the optimal shapes. Separately, 287,760 arbitrary marked subsets through three keys check the disjoint-charge bound without choosing the largest weights.

There are 6454 complete constructor controls against shape or original all-root optima, including rational scaling, geometric weights, isolated positive outcomes and zero-mass intervals; three empty inputs are checked separately. The all-root references use 1,357,542 root candidates. Twelve additional full vector constructions through 384 keys use three key runs and moderate or arbitrarily large exceptional key/gap weights. Every one matches an independent ordinary Knuth optimum and has its original tree cost, order and height checked. These twelve runs deliberately force the vector kernel: at these sizes the work selector would choose Knuth.

For the 384-key row with one 2^{192} key and one 2^{128} gap, the untrimmed sufficient height is 278; marking those two outcomes gives height 18, with remaining mass 767 and $\mu = 1$. The full forced kernel makes 12,489,698 finite run-entry queries, processes 524,018 singleton states, and stores 1,835,008 predecessor slots. This verifies the construction and height theorem, and does not establish a practical speed advantage at that size. A separate 1024-key geometric input checks iterative reconstruction of the complete rightmost-root chain against its endpoint-dominance optimum $2^{1025} - 1026$.

Twelve height-only plans with two exceptional outcomes and background keys in $\{1, 2, 3\}$ give heights 20, 23, 26 and 29 at sizes 1024, 4096, 16,384 and 65,536. Exceptional exponents 64/80 and 4096/4112 give the same budgets; no full construction is run at these sizes. A separate all-root audit checks 145,424 intervals across 1552 original or globally normalized inputs, using 2,551,952 root candidates and retaining minimum height among tied optimal costs. Rational and large weights are included. The records `trimmed_height_audit.json` and `trimmed_interval_audit.json` retain separate operation counters, reference scopes, and seven and three executed-source hashes, respectively.

Valuation-class records and sparse full-tree reconstruction. The record audit first checks every nonempty slot subset through height three: 135 subsets and 461 selected-key depth inequalities. A separate 6231-window control checks exact highest-valuation slot selection and earliest ties through height five, without scanning in the constructor.

All 680 binary mixed inputs through four keys, together with 60 further mixed, rational, geometric and run-structured inputs through 32 keys, give 1540 full constructions at prescribed heights. Every one matches both the dense prefix-vector optimum and an independent original all-root height recurrence. The audit checks all 226,280 prefix values and their attaining compressed predecessors, plus exact equality of the retained breakpoint sets at 6778 stage boundaries. At small heights, 53,824 literal slot selections provide a third prefix-value control. The original all-root reference uses 881,564 root candidates.

Eight larger complete constructions through 1024 keys cover a zero-weight middle run and genuinely mixed residual gaps. Their selected sufficient heights certify unrestricted optimality, and all match independent ordinary Knuth optima and direct output-tree cost, order and height checks. At 1024 keys, the zero-middle row uses height 25, implicit capacity 33,554,431, and 93,615 retained states across three stages. It makes 8,405,286 finite factor queries and 201,726,864 threshold-term evaluations; the repaired ancestor tree has 1034 vertices. These counts are distinct operation categories, rather than a wall-time comparison.

Three more full constructions keep 16 uniform keys while prescribing heights 16, 32 and 64. They all return cost 54, matching the closed minimum-internal-path-length optimum. At height 64 the implicit capacity is $2^{64} - 1$, while the constructor retains 15,120 prefix states, queries 302,820 destination rows, and constructs only 16 ancestor vertices. These rows check the absence of perfect-tree expansion; the integer complexity bound follows from the score-range proof, not from their measured sizes. The record `slot_records_audit.json` retains separate operation counts, reference scopes and seven executed-source hashes.

The marked-spectrum audit checks 60 inputs through five keys by enumerating 1428 slot selections, including rational marked weights and integer unmarked weights. Six further complete unrestricted mixed constructions through 192 keys place one exceptional key and one exceptional gap among three integer background runs. Gap exponents 64 and 512, paired with key exponents 80 and 528, all match independent ordinary Knuth optima. At 192 keys, both exponent choices select height 17 and retain the same 45,729 records across five stages; sparse repair constructs exactly 192 vertices. These equal counts are an observed control, while (41) proves the magnitude-independent bound. The separate `slot_marked_records_audit.json` retains six source hashes.

Exact exponential prefix-record support. The universal envelope in Lemma 165 has three independent controls. Earliest valuation-preserving embeddings are checked on all 32,902 nonempty slot subsets at heights one through four, covering 246,221 selected positions. Seventy-two arbitrary mixed rational inputs, including large exceptional keys, check 12,424 retained records against the envelope and compare 18,088 prefix values with the dense mixed recurrence; every complete output tree is independently checked. Finally, for each height $h = 1, \ldots, 6$ and every key count $1 \le n \le 2^h - 1$, geometric weights of base $h + 1$ attain the envelope at all prefixes: 2,667 stage sets and 97,155 dense states are checked, including $k > h$.

The positive mixed family in Theorem 167 is checked by a dense subsequence recurrence that retains every ending slot, independently of the compressed kernel. Across sizes 2 through 16, all 1,965,265 prefix values and attaining compressed predecessors agree. The same states check the popcount support test and the exact strict-increase locations of prefix maxima. All 135 complete record sets agree with the binomial formula. A separate enumeration checks 65,519 explicit subset-

of-scales witnesses. Through five keys, 208,376 literal slot selections provide another reference; 2055 literal BST shapes through eight keys verify the unique full optimum. Nine complete record-kernel trees through ten keys agree with that optimum.

At 16 keys, the middle prefix with eight keys has exactly 39,195 records; the final prefix has 65,520. The 135 measured stage counts also satisfy the summed identity in (168). Thirty further ascending, descending and shuffled magnitude-order diagnostics through 12 keys retain their counts without implying a universal upper bound. Formula evaluations at sizes 32 through 1024 are labeled separately; no exponential prefix enumeration is run at those sizes.

Six complete constructions of the same family by the existing affine solver reach 1024 keys. Every result matches its closed unique-chain cost and has all output links, order, depths and original mixed cost checked. Independent Knuth references are used only through 256 keys. At 1024 keys, the affine constructor uses exactly 2047 root candidates and returns the complete chain; the largest input weight has 10,242 bits. This checks the separation between explicit-record size and full-problem difficulty. The record `slot_record_barrier.json` retains exact counters, reference scopes, and eight executed-source hashes. No timing supports the theorem.

Implicit mixed prefix diagrams. The additive-diagram constructor is compared with a direct all-predecessor prefix recurrence on 72,880 entries, including infeasible positions. All 1392 complete small trees agree with an independent original all-root height recurrence and have every key, gap, order constraint, and depth checked. The inputs include all 680 binary mixed instances through four keys at two heights, plus rational and geometric controls through 24 keys. The direct prefix reference evaluates 486,336 predecessor candidates; these are reference operations, separate from constructor work.

Five complete geometric constructions use the general diagram solver at necessary height $h = n$, for $n = 8, 16, 24, 32, 48$. Each agrees with the closed unique-chain optimum and independent ordinary Knuth. At 48 keys the final function has 45,391 reachable diagram nodes, while its explicit record count is $2^{48} - 48$. Construction creates 3,069,517 nonterminal nodes, keeps 1,145,477 nodes reachable from all prefix roots, and makes 3,282,551 minimum calls, 2,290,986 addition calls, and 2,200,220 prefix calls. The large difference between final and intermediate sizes is retained explicitly. A 64-key scale run was interrupted and supplies no completed row or complexity evidence. The source-hashed record is `slot_dag_audit.json`; no timing is used.

A separate prefix-operation probe checks 108 signed functions with arbitrary infeasibility patterns at heights one through nine, comparing all 12,264 positions and checking the stated linear-in-hD operation bounds. The monotone-minimum family is checked for $m = 1, \ldots, 12$: every normalized middle-level node is counted, and the total is exactly $3 \cdot 2^m - 2$. There are 45,744 address controls of both operands and their minimum, exhaustive through $m = 6$ and sampled at the proved kink boundaries thereafter. The formula has the independent proof in Theorem 168.

Nine additional unrestricted mixed constructions use certified sufficient heights and match ordinary Knuth plus complete output-tree checks. They include random and near-uniform keys and gaps through 64 keys, and geometric keys with nonconstant, nonaffine gaps through 16 keys. The 64-key random row has height 11, 413 final nodes, and 187,957 total created nodes including its terminal; the near-uniform row has height 10, 230 final nodes, and 50,714 total nodes. At 16 keys the nonaffine geometric row has height 16, 1283 final nodes, and 199,887 total nodes. These complete controls establish correctness on the displayed inputs; the measured sizes supply no universal graph-size bound. The record `slot_dag_probe.json` retains input weights, operation counters, reference counts, and source hashes.

Actual-prefix size search and skyline traceback. A directed search uses exact dense mixed prefix values, independently normalizes every dyadic block, and maximizes the largest reachable diagram across all stages. It evaluates 204 inputs at $n = h = 6, 8, 10, 12, 14, 16$, using shuffled geometric keys, geometric gaps, and local coefficient or swap mutations. Every full optimum agrees with ordinary Knuth. The largest sizes found are respectively $58, 157, 310, 548, 1335, 2479$; the corresponding final sizes are $49, 81, 140, 311, 341, 1920$. Each winner has every prefix entry independently compared with the implicit constructor. Height n is sufficient on these inputs; the search does not certify that it is necessary. These finite sizes imply no growth exponent. The record is `slot_dag_size_search.json`.

A second search uses positive mixed inputs with necessary height n. For $A = 8(n+1)^2$, choose coefficients $\alpha_i, \beta_j \in \{1, \dots, n\}$ and set

$$p_i = \alpha_i A^{n-i}, \qquad q_j = \beta_j A^{\max(0,n-j-1)} \quad (0 \le i < n,\ 0 \le j \le n).$$

For a suffix beginning at i, put $r = n - i$ and let R be the mass of all its outcomes other than p_i. Then $R < 2nA^r/(A-1) + n$, so $nR/p_i < 3n^2/(A-1) < 1$. The right chain has suffix cost at most $p_i + nR$, while a different root incurs at least $2p_i$. Recursion forces the unique right chain and hence height n. This argument applies to every allowed coefficient mutation. At the same six sizes, 48 inputs are evaluated. The largest found sizes, also the winners' final sizes, are $48, 123, 266, 495, 829, 1283$. Every candidate has an independent Knuth optimum check; each winning complete skyline construction agrees with the closed chain cost and has all links, gaps, order constraints, and depths checked. The source-hashed record is `slot_dag_chain_search.json`.

The separate skyline audit checks 42,144 prefix values against direct all-predecessor minimization, 168,576 predecessor queries against literal candidate scans, and 185,856 endpoint-dominance inequalities. It includes all 680 binary mixed inputs through four keys and 24 rational inputs through 12 keys, totaling 704 complete all-root optimum and tree checks. Queries use zero, integer, and rational nonnegative gap prices, and include infeasible prefixes and the global endpoint sentinel. Six complete constructions on the unconstrained search winners agree with Knuth and all independently counted stage sizes. Their slot traceback makes exactly n^2 path steps at $h = n$ and creates zero diagram nodes, as predicted by Theorem 41. The record `slot_dag_skyline_audit.json` retains operation counts, executed source hashes, and the hash of its consumed search record.

Scalar-carry forward construction. The independent forward audit checks every one of 60,336 prefix entries against direct predecessor minimization, including infeasible positions. All 680 binary mixed inputs through four keys are checked at both the minimum feasible height and one larger height; 24 rational inputs through 12 keys add nonintegral controls. All 1384 complete trees agree with the original all-root height recurrence and have their links, order, depths, and original key/gap costs independently verified.

The same harness separately constructs the inclusive prefix-minimum diagram and checks 15,993 expanded carry contexts. For each actual non-pruned dyadic block it verifies that the inclusive-prefix minimum is the input block minimum, and that the normalized first-address value is exactly the relative carry, including infinity. Every executed context is covered, and distinct contexts have distinct input/prefix node pairs. This checks the earlier, weaker $O(hD^2)$ state bound by node-pair injection. The sharper reset-path proof of Theorem 43 applies to the same unchanged constructor and is audited separately below.

Twelve further full constructions reuse the retained winners of both size searches. Every stage has exactly the independently counted diagram size, and every complete optimum agrees with Knuth and output-tree checks. On the 16-key unconstrained winner, the scalar-carry constructor

creates 14,124 nodes versus 144,500 in the previous construction of that same input; it expands 3829 carry contexts. On the 16-key necessary-chain winner, the respective counts are 19,090 and 199,603 nodes, with 10,110 carry contexts. All twelve new constructions make zero calls to generic pointwise diagram minimum, addition, or prefix operations. The record `slot_dag_forward_audit.json` retains exact counters, nine executed source hashes, and hashes of both consumed search records.

A separate scale audit constructs positive mixed inputs from the strict-dominance chain family above, with random coefficients and necessary height $h = n$. Each full tree agrees with its closed unique-chain cost and ordinary Knuth; all links, depths, keys, and gaps are checked. Its exact counts are:

n	final nodes	all created nodes	history nodes	carry contexts
24	4928	106914	65358	58124
32	12410	358192	215203	197331
48	45391	1939134	1145477	1081655
64	111668	6355399	3722796	3567441

The all-created count includes the terminal; history counts nodes reachable from all retained prefix roots. At 64 keys there are also 38,536,277 first-address replacement visits, retained rather than hidden by the smaller final diagram. No timing or measured exponent supports the bound. The record `slot_dag_forward_scale.json` retains every original weight, eight source hashes, and all operation categories. These inputs are distinct from the earlier interrupted 64-key experiment.

Reset paths and exact generic sharpness. A separate audit checks the reset-path cover of Lemma 42 on 1416 updates, examining all 34,186 expanded states and 18,513 reset edges. For each update it reconstructs inherited and reset edges, checks at most one inherited child, checks that each parent input node determines a single reset seed, and verifies that the seed paths cover exactly the expanded states. It checks both the unweighted and depth-weighted bounds, every node-assembly and replacement count in (52), the recursive call identity, and the summary bounds. No generic pointwise minimum, addition, or prefix call is made.

There are 148 arbitrary-diagram controls, including signed rational and infinite entries, with four compact inputs at heights 16, 32, 64 and 128. They test the counting lemma only. Separately, 372 complete mixed trees are verified: all 168 binary inputs through three keys at two heights, 24 rational inputs through 16 keys, and the twelve retained size-search winners. The first 360 have original all-root height references; the winners have independent Knuth references and independently counted stage sizes. Every complete tree is checked against its original weights. The record `slot_dag_reset_audit.json` retains all counters, nine executed-source hashes, and both consumed search-record hashes.

The sharpness audit checks the exact closed formulas of Theorem 169 for $1 \le m \le 16$ and for $m \in \{24, 32, 48, 64\}$. It examines 14,700 expanded contexts and checks 7085 input values, exhaustively through $m = 6$ and with labeled samples thereafter. At $m = 64$, the 129-node input at height 128 causes exactly 6240 expanded contexts and 316,096 node-assembly calls, with the same number of replacement visits. Every row also checks the explicit endpoint-dominance violation. These are general-diagram operation controls, not complete OBST constructions. The record `slot_dag_reset_sharp.json` retains seven source hashes. The closed formulas supply the asymptotic tightness independently of these finite checks.

Punctured diagrams and complete mixed construction. The new representation is checked against all 60,336 direct prefix entries and 94,640 direct predecessor queries with integer and rational gap prices. All 1384 small complete trees agree with the original all-root height recurrence: every

binary input through four keys at two feasible heights, plus 24 rational inputs through 12 keys. Every output tree's original key/gap cost, order, links, and depths are checked.

On 96 arbitrary signed/rational/infinite diagrams, conversion agrees with independent dense canonicalization and all 6024 nonzero-address values. These inputs check representation and operation counts only. Twelve retained adversarial mixed inputs additionally have every prefix compared by canonical equality with conversion of the earlier ordinary-diagram constructor. Their complete trees agree with independent Knuth optima. Across these controls and the generic sharp family, 5456 updates check all 59,358 expanded states and 26,108 reset edges, exact reset-path coverage, $Z + T$ assembly calls, $2Z + 1$ recursive calls, and the summary bounds. The sharp family checks Corollary 170 through $m = 64$, including its exclusion from the OBST domain. At $m = 64$ its 128-node input requires 6240 expanded states and exactly 6303 assembly calls.

The sorting-free repair is separately checked on every one of 32,902 nonempty slot subsets through height four. Its complete output tree equals the retained sorting-based repair in every case; the ancestor traversal makes exactly two visits per collected vertex. This checks the replacement of sorting by traversal independently of the optimizer. The record `slot_punctured_audit.json` retains all counts, ten source hashes, and both consumed size-search hashes.

A separate full-construction audit uses the identical necessary-height mixed chain inputs of the earlier scale audit. Each constructed tree matches the unique-chain formula, independent Knuth, and all original weights. Exact counts for the punctured representation are:

$n = h$	final nodes	all created nodes	assembly calls
24	3054	42092	58424
32	7806	143956	197859
48	29143	795874	1082831
64	72235	2636579	3569521

The same 64-key input previously created 6,355,399 ordinary nodes and performed 38,536,341 assembly calls and 38,536,277 first-address replacement visits, including the initial ordinary point diagram in the assembly count. The new constructor performs no first-address replacements. These are matched exact operation counts; the theorem supplies the bound.

Four additional arbitrary mixed integer inputs at 32, 64, 128 and 256 keys use the proved sufficient height selector and agree with independent Knuth and complete-tree checks. The selected heights are 11, 13, 14 and 16. At 256 keys the final punctured diagram has 460 nodes; complete construction creates 73,226 nodes and makes 114,633 assembly calls. All intermediate work is retained, and these measurements assert no all-stage size bound. The record `slot_punctured_scale.json` retains every original weight, six source hashes, and the consumed earlier scale-record hash.

Actual-prefix cubic and shared-history quartic witnesses. The audit for Theorem 171 first compares every greedy ending-slot reward with an independent dense subsequence recurrence for $2 \le n \le 12$, giving 89,749 exact entry comparisons. Eight complete constructions at $n = 12, 13, 16, 17, 24, 25, 32, 48$ then check all three entries of every witness at every prefix. There are 507,376 witnessed nodes and 1,522,128 entry comparisons. Each witness also checks the first and last nonzero coefficient positions and the coefficient identifying s. Actual canonical node identities are distinct across all prefix roots, so the history test includes full cross-stage sharing. Odd sizes check the even probe offset $Q = 2\lceil n/2 \rceil$ separately from $Q = n$.

Every full output agrees with independent Knuth, the closed chain cost, and original-weight tree checks; all 187 suffix dominance certificates hold. The finite node counts and proved lower bounds are:

n	final nodes	final lower bound	shared history	history lower bound
12	287	56	2044	434
13	379	84	2912	714
16	767	165	7242	1815
17	937	220	9429	2585
24	3054	969	42092	16473
25	3488	1140	50176	20235
32	7806	2600	143956	60450
48	29143	11480	795874	404670

The record `slot_punctured_barrier.json` retains every stage count, all operation counts and seven source hashes. Rows from 64 through 4096 keys evaluate only the proved formulas; they are not enumerations or full constructions. The lower-bound proof is independent of these measurements.

Positive affine block sharing. The audit for Theorem 173 checks all 43,941 unordered quadratic exponents at selected sizes through $n = 256$ for uniqueness. It then compares 18,213 greedy scores with independent dense subsequence values at $2 \le n \le 10$. Five complete constructions at $n = 8, 12, 13, 16, 17$ check every witness's three entries against the actual prefix function: 5567 distinct affine invariants and 16,701 exact entry comparisons. Every invariant is also checked after two nontrivial positive affine transformations, giving 11,134 exact rational controls. Distinctness is checked across all prefix stages, and the coefficient tests verify linear independence and recovery of r, s, k.

All five original-weight trees agree with ordinary Knuth and the closed chain cost, with 66 strict suffix certificates. The largest normalized weight has 10,322, 38,740, 50,247, 98,499 and 119,841 bits in the five rows, respectively. These finite bit lengths are reported separately from arithmetic-operation counts. The record `slot_affine_barrier.json` retains each exponent sequence, stage counts, exact operation counts and eight source hashes. The existing additive solver supplies the tested function values; no affine-normalized solver is claimed or required for the equivalence-class lower bound.

Lazy search, valid bounds, and forced exponential states. The audit for Lemma 174 exhausts the 680 binary mixed inputs through four keys and adds 48 seeded rational inputs at $n = 5, 8, 12$. At two heights per input it checks 55,744 bound entries against dense exact prefix values and independently sorted dot products. It checks all 14,245 evaluated prefix states against fresh all-predecessor recurrences and all 1456 height-constrained optima against the original all-root recurrence. Another 2184 complete constructions use budgets zero, seventeen, and the default $8n^2$. All original-weight tree costs match unrestricted Knuth. Of these, 2174 use fallback and ten finish the lazy search; every exhaustion occurs at exactly one counted operation beyond its budget. The record `slot_lazy_audit.json` retains these counts, seed and eight source hashes.

The matched scale audit retains 22 complete constructions with independent Knuth controls. Five uniform rows at $n = 32, 64, 128, 256, 512$ finish the lazy search, expanding exactly n states each. Their counted search work is 5290, 12,756, 30,263, 87,117 and 199,099 operations, respectively. All ten bounded and binary mixed rows at those sizes reach the default budget and then construct the optimum with Knuth; their unsuccessful search work is retained. The chain family uses an explicit diagnostic budget of two million search operations. Its completed and budget-exhausted rows are:

n	Necessary h	Expanded states	Search work	Completion route
6	6	67	4221	Lazy
8	8	382	32,013	Lazy
10	10	2037	211,825	Lazy
12	12	10,216	1,267,315	Lazy
14	14	12,587	2,000,001	Knuth fallback
16	16	8689	2,000,001	Knuth fallback
18	18	6554	2,000,001	Knuth fallback

The record `slot_lazy_scale.json` retains every original input, height, budget, search and fallback counter, separate reference counter, seed and eight source hashes. These finite counts are not a fitted exponent or a speed claim. Uniform trees already have a linear constructor.

A separate unbudgeted audit for Theorem 175 completes $n = 6, 8, 10, 12$, checking all 14, 62, 254 and 1022 distinct states in (184). Their canonical paths contain 14, 62, 254 and 1022 distinct edges. Every edge is checked for skyline membership, feasibility, strict comparison of its candidate bound with the exact parent optimum, and presence of its child in the completed search cache. The full runs cache 68, 383, 2038 and 10,217 exact states, including base states. Post-search bound comparisons add operations to this audit's counters; the scale table above reports construction search work alone. The record `slot_lazy_barrier.json` retains all counters and eight source hashes. Its larger rows, through $n = 256$, evaluate the proved subset-count formula only. The theorem follows from the explicit paths and strict dominance, independently of these finite checks.

21 Unimodal domination and certified-envelope audit

The audit exhausts 920 unimodal words over $\{0, 1, 2\}$ through eight keys and checks all 136,802 nonempty subsequences for componentwise domination by the constructed contiguous block. Its independent definition rejects 8920 nonunimodal words. Another 1589 nonnegative reward-template controls, from 12,864 literal subsequences, check the optimization identity without assuming ruler rewards.

On 236 mixed inputs through 32 keys, including monotone, constant, isolated peak, geometric, irregular unimodal and rational weights, the original all-root recurrence checks 31,438 interval values and contiguous witnesses, 560 complete above-capacity support sets, and 24,032 densities. All 99,398 eligible small square controls satisfy $2s - 1$. The original recurrence uses 565,705 root candidates and 65,081 states.

The arbitrary-input audit includes every binary word at nine and ten keys, plus 100 random mixed inputs through 32 keys, totaling 1636 inputs. It independently checks 101,300 interval-unimodality decisions, 200,892 upper-cost inequalities, 91,143 certified exact interval values, and 130,408 envelope densities. The original recurrence checks 38,221 exact certified densities and 4988 certified support sets, using 2,345,440 root candidates and 435,040 states. The two signed-correction examples have eight literal score controls from 253 subsequences, separate from those recurrence checks.

Fourteen larger sparse constructions reach 65,536 keys. They have 1168 sampled literal ruler-window controls and 2780 constructed local-square controls; five single-key rows also match the sharp closed-form count. They do not have original all-root layer references at those sizes. At $n = 65,536$ both fixtures perform 917,538 ruler-window updates and 983,026 range-tree build-node operations. The isolated-peak input outputs 65,294 core entries across all layers, and the irregular rational mountain outputs 1201. All entries are constructed without scanning interval pairs.

The record is `unimodal_above_audit.json`. It retains exact operation counts, both signed-correction witnesses, verification scopes, and the final executed-source SHA-256 manifest. The support and work bounds follow from contiguous domination, the ruler recurrence and visibility geometry.

22 Valley domination, implicit queries, and dense-square audit

The independent audit exhausts 920 valley words over $\{0, 1, 2\}$ through eight keys. It checks all 136,802 nonempty subsequences for componentwise domination by the selected end blocks, and 6198 arbitrary nonnegative reward vectors. An independent definition rejects 8920 nonvalley words.

On 256 mixed inputs through 64 keys, the original all-root recurrence checks 56,076 above-capacity interval values and their two-block witnesses. Inputs include monotone and constant words, geometric and irregular valleys, rational weights, and nonzero gap weights. The original recurrence uses 1,396,993 root candidates and 113,799 states. The explicit dense family and its normalized rational version have 672 square cells checked against that recurrence, at sides four, eight, and sixteen. Each square also has the proved exact support count $s^2/2$.

Fourteen larger implicit constructions reach 65,536 keys. Their checks comprise 896 sampled literal largest-key selections, 896 literal ruler-prefix sums at arbitrary lengths, and 640 sampled dense-formula cells. These are separate from the original all-root controls; no original all-root layer is computed at the large sizes. At $n = 65,536$, each construction performs 1,835,076 ruler-window updates and stores 1,966,150 window entries across the forward and reversed arrays, including their zero bases. No interval table or dense core is constructed.

The record `valley_above_audit.json` retains operation counts and the executed-source SHA-256 manifest. The theoretical work bounds follow from the two-wing binary search and ruler-prefix recurrence; the core lower bound follows from (163) and its residue count.

23 Valley support charges and sparse-core construction audit

The audit uses 1513 small inputs: all 1461 ternary valley words through nine keys, 48 varied mixed inputs through 32 keys, and four integer or normalized rational dense-family inputs. Literal sorted sums check 73,082 auxiliary densities, 32,810 two-threshold formulas, and 4267 complete auxiliary support sets. The same number of constructed ruler support sets agree with dense evaluation of the independently audited query representation. Every eligible small square gives a boundary-mass control, totaling 198,321. Integer inputs additionally check the support count against that mass.

Original all-root mixed layers, separately from those query controls, check 49,278 densities and 2644 complete original support sets on 972 inputs. They use 1,006,805 root candidates and 163,435 states. The nonvalley word p^+ in Section 5.23 has a separate original control, with 495 root candidates and 135 states, confirming original density one and auxiliary density zero.

Twelve larger constructions through 16,384 keys enumerate all above-capacity cores on binary end blocks and bounded integer valleys. Their 3264 sampled literal density controls are kept separate from the original all-root controls. At the largest size the binary input emits 4084 original entries from 8178 auxiliary events; the bounded valley emits 16,477 from 32,995. Both preprocessings perform 393,276 ruler-window updates. Five complete dense-family constructions also check 5456 cells against (163), through side 64. The record `valley_core_audit.json` retains all operation counts, scopes, the nonvalley witness, and executed-source hashes. The support and work bounds follow from the charge and event-enumeration proofs.

24 Periodic phase construction and full-tree audit

The independent audit checks 1042 small inputs: all mixed binary vectors through four keys, every binary key/gap period through length three at several complete and incomplete repetition lengths, and seeded geometric, zero, rational, and final-gap-defect inputs through 48 keys. Literal period enumeration checks every detected joint period. Independent all-root tables check 62,376 original interval values and 54,672 leftmost roots. Extended all-root tables additionally check all 38,756 synthetic phase states used in the cyclic cancellation, including intervals crossing the input endpoint. These references use 1,993,994 root candidates and 154,391 states. Separate ordered-tree enumeration evaluates 7882 shapes on the small exhaustive cases.

Twenty larger matched instances through 512 keys use an independent ordinary Knuth recurrence, which examines 1,744,227 candidates. Every reconstructed tree is traversed independently, checking inorder placement, all child links, all key and gap depths, and weighted cost. Sixteen further complete constructions reach 262,144 keys, with periods one, three, and five and both zero and positive gap patterns. At that size the period-five mixed input uses exactly 2,621,435 root candidates, and the period-three input uses 1,572,861, as predicted by (76). The all-zero case reconstructs the full leftmost-optimal chain iteratively. These large rows verify complete construction, exact operation counts, and independently evaluated tree costs; they do not claim an independent quadratic optimality reference at that size.

The result record retains optimal costs, reconstruction counts, verification scopes, and executed-source hashes. No timing or fitted exponent is used. The complexity and optimality claims follow from the exact quotient (74) and cyclic amortization (76), both of which include the synthetic phases.

25 Boundary extension and full mixed-tree audit

The implementation first checks Wilber's LWS primitive against a fresh dense recurrence: 768 instances through 64 vertices, with signed finite row and column offsets and quadratic separation costs, use 549,120 literal arc candidates. All values and leftmost predecessors agree. The implementation follows the two-search batching algorithm in Wilber's Figure 1. Invalid cells internal to SMAWK use an ordered tag above all finite values; valid arc queries remain finite. Each batch's comparisons therefore equal those of a sufficiently large finite completion. The implementation records batch widths and checks that their sum is at most three times the number of vertices; this is the potential argument from Wilber's linear bound.

The full mixed audit checks 980 small inputs: all binary mixed vectors through four keys and seeded periodic-middle inputs through 27 keys, including rational weights and boundary weights as large as 10^{40}. Explicit middle selection and automatic selection are checked separately. Independent all-root tables use 351,256 candidates and 49,492 states, verifying 57,416 original interval values and 44,230 stored root witnesses. Another 6944 rightmost-root controls test the new-column reduction directly against the original tables, independently of the periodic solver. Literal substring enumeration checks 3324 longest-middle decisions using 162,564 candidate substrings. Small ordered-tree enumeration evaluates 7880 shapes. Every reconstructed tree is independently traversed to check ordering, links, key and gap depths, and weighted cost.

Sixteen matched inputs through 512 keys use the ordinary uncompressed Knuth recurrence, examining 1,847,784 candidates. Twelve larger complete automatic constructions reach 65,537 keys, with periods one, three, and five and one or seven arbitrary fringe keys. At that size, the one-key right fringe and period-one middle have exactly 65,537 new boundary states, 131,071 central root candidates, and 1,309,476 valid LWS transition queries. With period five and seven fringe keys, the corresponding counts are 458,738, 655,295, and 8,264,895. The automatic scan performs 1,113,976 signature comparisons on each of these largest inputs. These are operation counts; central root candidates and LWS transition queries are different operations.

The large rows verify complete construction and independently evaluated tree costs; they have no separate quadratic optimality oracle. The work bound comes from Lemma 56 and the exact state identity (80). The result record retains the seed, costs, operation counts, verification scopes, and six executed-source hashes. No timing or fitted exponent enters the theorem.

26 Affine translation and finite-domain construction audit

Before running the quotient solver, the harness checks the shape-cost identity on 6250 literal ordered trees through seven keys. The transformations include positive scales below and above one and offsets of both signs. The full audit then checks 1063 small mixed inputs: all binary key/gap vectors through four keys, seeded affine inputs through 32 keys, and 21 finite inputs whose hypothetical continuations develop negative weights. Explicit witnesses include periods close to the input length, testing the terminal diagonals as well as the complete phase cycles.

Independent all-root tables examine 640,661 candidates and 74,877 states. They check 84,584 original interval values, all 74,877 leftmost roots, and 28,534 stored phase states. Ordered-tree enumeration separately checks 7882 small optimal costs. Recognition has an independent reference: solving the original two-variable linear equations checks 8644 period-feasibility decisions, and 860 automatic inputs verify the smallest admissible period. This reference does not use the production adjacent-sum test. Another 2916 checks validate returned parameter witnesses directly.

Sixteen matched inputs through 512 keys use the ordinary uncompressed Knuth recurrence, examining 1,393,110 candidates. Nine larger arithmetic-drift constructions reach 65,537 keys, with periods one, three, and five. At that size the period-one construction examines exactly 131,073 root candidates; the period-five case examines 655,325 phase candidates and 18 terminal candidates. The candidate bounds and exact state count are checked against (85) and (86). Two further exact geometric constructions reach 4095 keys; their integers grow with the input length, so their counts concern arithmetic work.

Every output is independently traversed to check key order, child links, key and gap depths, and weighted cost. The eleven largest construction rows have no separate quadratic optimality reference; their optimality guarantee is Theorem 60. The result record retains the seed, operation counts, exact costs, per-row reference scopes, and four source hashes. No timing, fitted exponent, or infinite positive extension enters the proof.

References

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